Academic submission

MedChain

Blockchain-Based Patient Consent Management and Secure Medical History Sharing

Algorithm 08 — Patient Consent Management Contract

Implementation: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Web Crypto API (SHA-256), localStorage persistence.
Evidence in this document was captured from the running application on 06 Nov 2026, 07:11:31. Every event ID, transaction ID and hash shown is taken from that run; nothing is typed in by hand.

The blockchain network, smart-contract deployment and credential signatures are simulated in the browser, as allowed for the initial demonstration. The access-decision logic and the SHA-256 hash chain are real and execute as described. All patient data is synthetic.

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1. Objective

MedChain demonstrates how a smart contract can control access to a patient's medical history when it is shared between hospitals. A record is released only when **all** of these hold:

- the requesting healthcare professional is authenticated, verified and authorized (Rule 1);
- the patient has granted consent to that hospital or professional, and the requested records are within the consented scope (Rule 2);
- the request's purpose matches the consent, and the consent is active, not expired and not revoked (Rule 3);
- and the source hospital approves the request.

Every action in the process is written to a SHA-256 hash-chained ledger that is visible live on the right side of the screen.

Patient grants consent → Physician authenticates → History requested → Source hospital receives and verifies → Algorithm 08: R1, R2, R3 → Hospital responds → Access granted / denied → Records released → Ledger

2. Running the application

```
npm install
npm run dev
# open http://localhost:3000 (window at least 1200 px wide so the live ledger stays on the right)
```

The application has four screens, matching the assignment:

Screen	Purpose
Request Workflow	The complete process on one page in six numbered steps. Each step states who acts; a "Sign in as …" button appears when the next step needs a different person.
Patient Portal	Profile, record categories, grant() and revoke() of consent, active permissions, requests for the patient's records with decisions, consent and access history, demo clock (for the expiry test).
Smart Contract Studio	Contract ID, version and SHA-256 hash; the three fixed rules; the pseudocode and the actual TypeScript executed; execution history; simulated deployment receipt.
Transaction Ledger	All events with search and filters, transaction history, expandable details with hash inspection, Verify Ledger, tampering demonstration, CSV export, Reset Demo Data.

Layout. Left: navigation. Centre: the workspace. Right: the **live transaction ledger**, which shows each event as soon as it has been hashed and saved: event ID, transaction ID, request ID, event type, actor and role, requesting and source hospitals, patient, record categories, purpose, timestamp, status, previous and current hash. The newest event is highlighted, and each event expands to its full details.

3. Users, roles and sample data

Demo user	Role	Hospital	Used for
Ayesha Khan	Patient	—	Grants and revokes consent; views requests and decisions
Dr. Ahmed Raza	Physician	H1 City Hospital	Authorized requester
Dr. Victor Hale	Physician	H5 External Hospital (not registered)	Unauthorized requester for the Rule 1 test
Sara Malik	Hospital Records Officer	H2 General Hospital	Receives, verifies, runs the contract, approves or rejects, releases
Imran Qureshi	Hospital Administrator	H3 Medical Centre	Creates and deploys contract versions; cannot override rules
Grace Mensah	Auditor	—	Verifies the ledger

Patient: Ayesha Khan (PAT-001) with synthetic LAB, ECG and MRI records held at H2 General Hospital.

Hospitals: H1 City Hospital, H2 General Hospital, H3 Medical Centre, H4 Central Hospital (registered and active); H5 External Hospital (not registered, unauthorized).

Initial consent (CON-0001): authorized hospital City Hospital (H1); permitted records LAB and ECG; MRI not permitted; purpose TREATMENT; expiry 31 December 2026; status ACTIVE.

Why these users: one user per role named in the assignment, plus one unauthorized physician so that Rule 1 can be shown failing on real data rather than on a hard-coded flag.

4. Algorithm 08 — Patient Consent Management Contract

The algorithm is one TypeScript module, `src/lib/contract.ts`, with the three functions of the lecture definition. Every screen calls these functions through the central store; there is no second copy of the decision logic. The functions are pure: the same inputs always give the same decision.

Variables (as in the lecture): `p` = patient DID; `g` = grantee DID (a hospital or a designated professional); `scope` = record types allowed; `type` = record type requested; `purpose` = TREATMENT | RESEARCH; `exp` = expiry; `now` = block time; `guardian(p)` = the patient's legal guardian.

4.1 Mapping to the lecture's Algorithm 08

#	Lecture line	Implementation	Notes
1	`grant(p, g, scope, purpose, exp)`	`grant()` in contract.ts, called by the Patient Portal	Same five inputs. Also requires a registered patient DID, a non-empty scope and an expiry in the future within the policy limit (365 days).
2	`require caller ∈ {p, guardian(p)}`	`requirePatientOrGuardian()` inside `grant()` and `revoke()`	The signed-in caller's DID must equal the patient's DID or the patient's guardian DID. Ayesha is an adult, so she has no guardian in the demo data.
3	`consents[p][g] ← {scope, purpose, exp}`	Consent stored with `patientDid` and `grantee.did`; the key `consents[p][g]` is shown on screen and in the ledger	A new grant for the same (p, g) replaces the previous entry, so there is one active entry per pair.
4	`check(p, g, type): type ∈ scope ∧ now < exp`	`check()` → `evaluateRule(1..3)`	`type ∈ scope` is Rule 2; `now < exp` is Rule 3. The assignment's fixed rules add identity checks (Rule 1), and the purpose match and revocation status (Rule 3).
5	`revoke(p, g): delete consents[p][g]; emit`	`revoke()`; CONSENT_REVOKED is recorded	The entry is marked REVOKED instead of being physically deleted. For access the effect is identical (check() can never pass again), but the assignment's Rule 3 must report "revoked", and the audit trail must keep the record.
6	every access path calls check() first	`check()` runs at the contract step, **and again at the moment of release** (ACCESS_CHECK_AT_RELEASE)	A consent revoked or expired after approval still blocks the release (demonstrated in §8.4).
—	now = block time	Browser clock (plus the demo clock for the expiry test)	Simulated: there are no real blocks.
—	Sample input: scope {LAB, ECG}, TREATMENT, exp 2026-12-31; check(..., MRI) → false	Seed consent CON-0001 has exactly these values	The MRI request is denied by Rule 2 (§7).

4.2 grant(p, g, scope, purpose, exp)

```
require caller == p  or  caller == guardian(p)          (line 2)
require p has a registered DID, scope is not empty
require now < exp <= now + maxConsentDays
consents[p][g] = {scope, purpose, exp, ACTIVE}           (line 3; previous entry for (p, g) superseded)
emit CONSENT_GRANTED
```

Reasoning. Only the patient, or someone legally acting for them, can share their data. Keeping one entry per (p, g) means check() never has to choose between conflicting consents, and the expiry limit stops open-ended sharing.

4.3 check(p, g, type)

```
consent = consents[p][g]      (newest ACTIVE entry for the professional's DID or their hospital's DID; else the
latest one)
if not R1(g, processor)                       → DENY (Rule 1)
if not R2(consent, g, type)                   → DENY (Rule 2)   type ∈ scope
if not R3(consent, request.purpose, now)      → DENY (Rule 3)   now < exp, purpose matches, not revoked
→ APPROVE
```

Why this order. Identity is checked first so that an unauthorized party is stopped before the contract reads the patient's consent at all. Scope (R2) is checked before validity (R3) because a request for records the patient never consented to is best explained as out of scope. Evaluation stops at the first failure, like `require()` in a smart contract.

Why the latest entry is used when none is active. If a consent was revoked or has expired, Rule 3 can then say exactly that ("consent CON-0001 is REVOKED") instead of "no consent found".

4.4 revoke(p, g) and line 6 (every access path calls check() first)

```
revoke(p, g):
  require caller == p  or  caller == guardian(p)
  consents[p][g].status = REVOKED                      (line 5; equivalent to delete for access)
  emit CONSENT_REVOKED

releaseRecords(request):                               (line 6)
  require contract decision == APPROVE and hospital approved
  require check(p, g, type) == APPROVE                 (re-run at the moment of access)
  release only records whose type was requested and is in scope
```

Reasoning. Consent can change between the contract decision and the moment the records leave the hospital. Re-running check() at release means the patient's latest decision always wins.

4.5 The three fixed rules

Rule	Condition	Checks (actual inputs shown on screen)	Why it exists
R1 Identity and hospital authorization	`user.authenticated && user.credentialVerified && hospital.status == ACTIVE && role.hasPermission`	Requester signed in; requester credential verified (trusted issuer, unaltered, in date, not revoked); requesting hospital ACTIVE; role Physician has request:create; the processing records officer is signed in, verified, has request:process and works at the source hospital; source hospital ACTIVE.	Only known, verified professionals at registered hospitals may request or process patient data.
R2 Patient consent and record scope	`consent != null && grantee IN consent.grantees && requestedRecordType IN consent.allowedScope`	A consent exists for the requesting hospital or the designated professional; every requested category is in the consented scope.	The patient decides who sees which records (data minimisation).
R3 Purpose, expiry and revocation	`request.purpose == consent.purpose && consent.status == ACTIVE && currentTime < consent.expiry`	Purpose matches; consent ACTIVE (not revoked or superseded); current time before expiry.	Consent is purpose-bound and time-limited, and the patient can withdraw it at any time.

4.6 No override, and release only after hospital approval

- A failed rule records CONTRACT_DENIED and ACCESS_DENIED and ends the request. The screens then show no Approve or Release action, and the store itself refuses approval unless the request's status is CONTRACT_APPROVED. No role, including the Hospital Administrator, has an override function.

- The baseline rules are fixed. Contract Studio validation rejects any new version whose rule text differs from the fixed rules.
- Release requires both: check() returned APPROVE **and** the source hospital approved. Only records that are both requested and inside the consent are released.

5. Hash-chained ledger — design and reasoning

Every meaningful action is written through one function, `recordLedgerEvent()`, inside a single queued commit (`src/lib/store.tsx`):

1. The action changes the application state and records its events with the actual actor, role, hospital, patient and request.
2. Each new event gets an event ID; events of the same request reuse the request's transaction ID.
3. Each event is hashed in order: **`currentHash = SHA256(previousHash + canonicalEventPayload)`**, using the browser's Web Crypto API.
4. The new state is saved to localStorage. Only then does any screen, including the live ledger, update. If saving fails, the action is rejected and reported.

Design choice	Reason
Genesis previousHash = 64 zeros	The first event has nothing before it; a fixed starting value makes the chain verifiable from the start.
Canonical payload (JSON with sorted keys, hash fields excluded)	The same event always produces the same text, and so the same hash, on any machine.
Sequential processing queue	Two quick actions can never compute their hashes from the same previous hash (no race in the chain).
Hash before display, persist before display	No event is ever shown as recorded unless it has been hashed and saved.
Shared state across browser tabs	A write always builds on the newest saved ledger, so one tab cannot overwrite events written by another.
Verification recomputes every hash	Changing any stored field changes that event's recomputed hash, so verification reports the exact event that was altered.

Fields of every event: Event ID, Transaction ID, Request ID, timestamp, actor, actor role, hospital, event type, execution result, explanation, contract version, patient and request context (patient, requesting hospital, source hospital, categories, purpose, reason), previousHash, currentHash.

Event types used: AUTHENTICATION_VERIFIED, CREDENTIAL_VERIFIED, HOSPITAL_VERIFIED, CONSENT_GRANTED, CONSENT_REVOKED, REQUEST_CREATED, REQUEST_RECEIVED, REQUEST_REVIEW_STARTED, HOSPITAL_VERIFICATION_COMPLETED, CONTRACT_EXECUTION_STARTED, RULE_1_EVALUATED, RULE_2_EVALUATED, RULE_3_EVALUATED, CONTRACT_APPROVED, CONTRACT_DENIED, HOSPITAL_APPROVED, HOSPITAL_REJECTED, ACCESS_GRANTED, ACCESS_DENIED, MEDICAL_RECORD_RELEASED, plus LEDGER_VERIFIED and the genesis events.

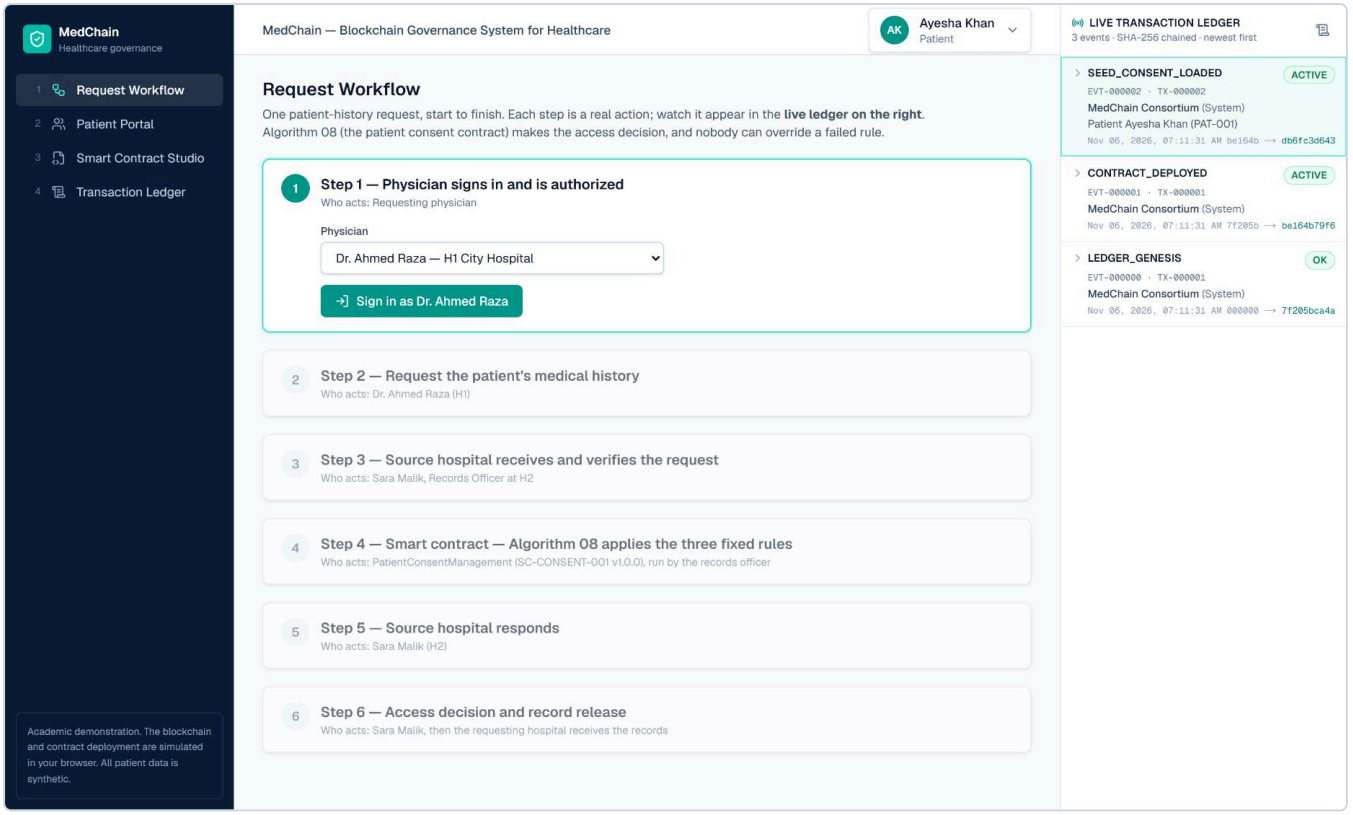


Figure 1. Starting point after Reset Demo Data: three genesis events in the live ledger (right). The first has previousHash 000000…

Genesis events

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000000 TX-000001	**LEDGER_GENESIS** Genesis event. Its previousHash is 64 zeros; every later event links to the one before it.	MedChain Consortium System	OK	000000000000… ↓ **7f205bca4af8…**
EVT-000001 TX-000001	**CONTRACT_DEPLOYED** PatientConsentManagement (SC-CONSENT-001) v1.0.0 deployed with the three fixed rules (simulated deployment).	MedChain Consortium System	ACTIVE	7f205bca4af8… ↓ **be164b79f68b…**
EVT-000002 TX-000002	**SEED_CONSENT_LOADED** Initial demonstration data (not a user action): Ayesha Khan (PAT-001) allows H1 City Hospital to access LAB and ECG for Treatment until 2026-12-31.	MedChain Consortium System	ACTIVE	be164b79f68b… ↓ **db6fc3d643d9…**

6. Case 1 — successful ECG request (approved)

Dr. Ahmed Raza (City Hospital, H1) requests Ayesha Khan's ECG from General Hospital (H2) for treatment. The consent allows LAB and ECG for H1, so all rules should pass. Request **REQ-2026-0001**, transaction **TX-000004**. Each step below shows the screen at that moment, with the live ledger on the right, and the events that step wrote.

Step 1 — Physician signs in and is authorized

> **Assignment requirement:** A healthcare professional logs in; the system verifies identity, role, credentials and hospital registration.

What the user does: Selects Dr. Ahmed Raza and clicks *Sign in*.

What the system checks (logic): The credential is verified (issuer is a trusted consortium hospital, proof hash matches the credential content, not expired, not revoked), the hospital's registry status is checked, and the role's permissions are read.

Why: These are the facts Rule 1 will rely on, so they are established and recorded when the user signs in.

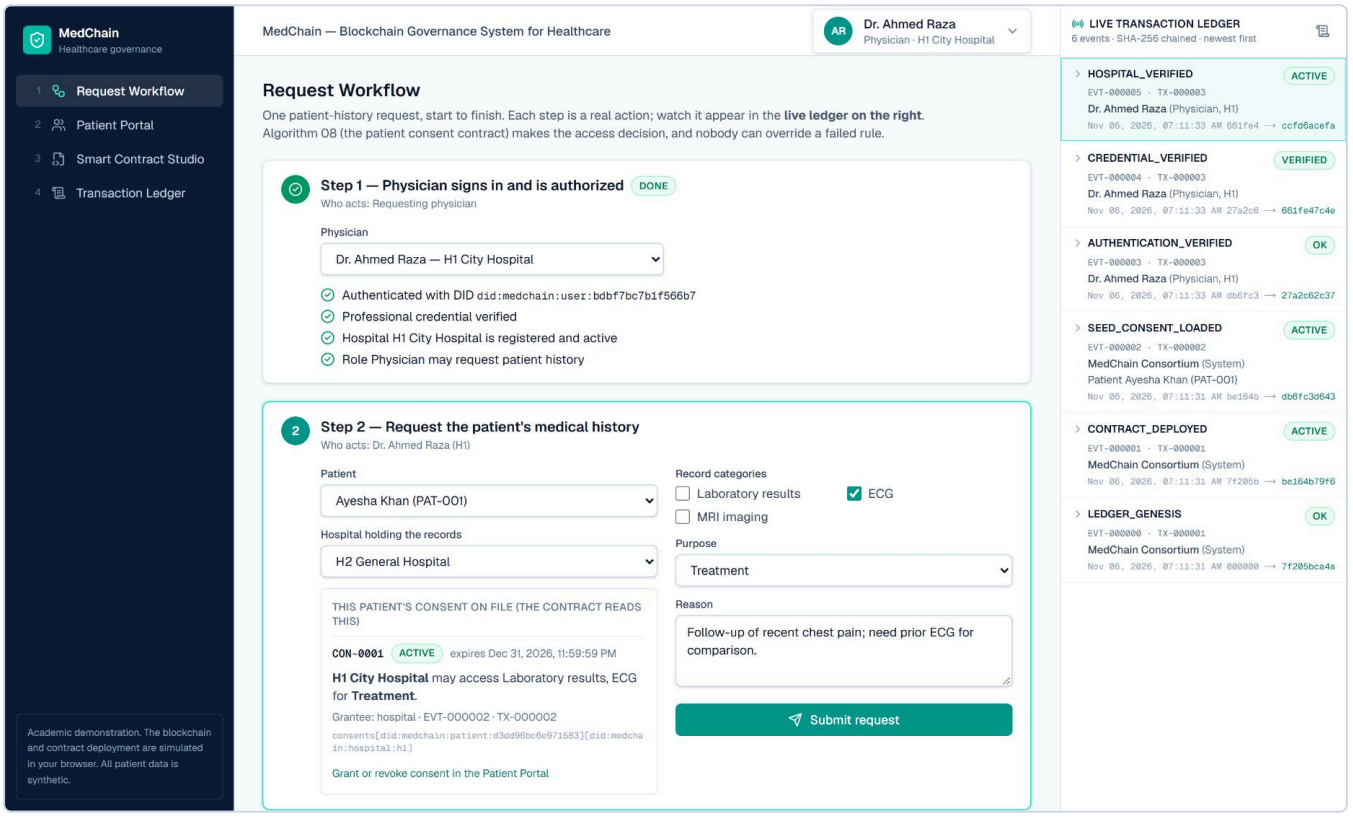


Figure 2. Step 1 complete: four green checks; three events appear at the top of the live ledger.

Live ledger: events written by this step (3)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000003 TX-000003	**AUTHENTICATION_VERIFIED** Dr. Ahmed Raza signed in as Physician using DID did:medchain:user:bdbf7bc7b1f566b7 (demo authentication).	Dr. Ahmed Raza Physician, H1	OK	db6fc3d643d9… ↓ **27a2c62c379e…**
EVT-000004 TX-000003	**CREDENTIAL_VERIFIED** Credential urn:vc:medchain:u-phy-h1 verified: trusted issuer, intact, in date, not revoked.	Dr. Ahmed Raza Physician, H1	VERIFIED	27a2c62c379e… ↓ **661fe47c4efd…**

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000005 TX-000003	**HOSPITAL_VERIFIED** H1 City Hospital is registered and active in the consortium registry.	Dr. Ahmed Raza Physician, H1	ACTIVE	661fe47c4efd… ↓ **ccfd6acefa54…**

Step 2 — Request the patient's medical history

> **Assignment requirement:** The physician selects a patient, the medical-history categories, the purpose and reason, and submits to the source hospital; a unique request ID and transaction reference are generated.

What the user does: Chooses Ayesha Khan, ECG, Treatment, a reason and source hospital H2, then clicks *Submit request*. The patient's consent on file is shown next to the form.

What the system checks (logic): Only a signed-in user whose role has request:create can submit. The request is placed in H2's inbox in the same saved change that records REQUEST_CREATED.

Why: The transaction ID ties every later event of this request together in the ledger.

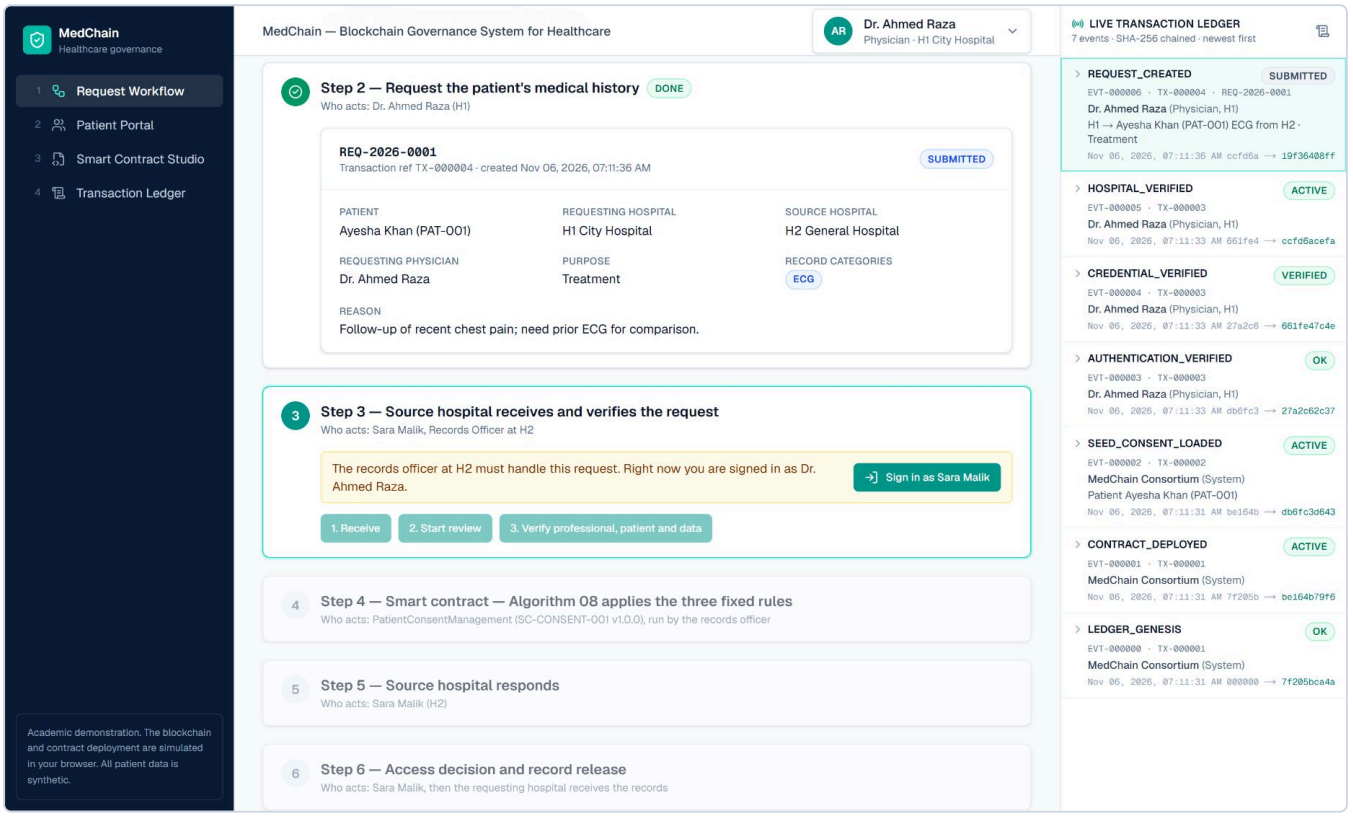


Figure 3. Request REQ-2026-0001 (transaction TX-000004) created. The live ledger shows who requested which patient's ECG, from which hospital and why.

Live ledger: events written by this step (1)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000006 TX-000004 REQ-2026-0001	**REQUEST_CREATED** Dr. Ahmed Raza (H1) requested Ayesha Khan's ECG from H2 for Treatment. Reason: Follow-up of recent chest pain; need prior ECG for comparison.	Dr. Ahmed Raza Physician, H1	SUBMITTED	ccfd6acefa54… ↓ **19f36408ffbb…**

Step 3 — Source hospital receives and verifies the request

> **Assignment requirement:** The source hospital receives the request; an authorized Records Officer reviews it and verifies the patient details, requesting hospital, physician and requested records.

What the user does: Signs in as Sara Malik (Records Officer, H2), then clicks *Receive*, *Start review* and *Verify*.

What the system checks (logic): Verification checks: patient reference (DID registered), requesting hospital status, requesting physician's credential, whether H2 holds the requested records, and whether a usable consent is on file.

Why: This is the hospital's own check. It cannot approve anything by itself; the contract still decides.

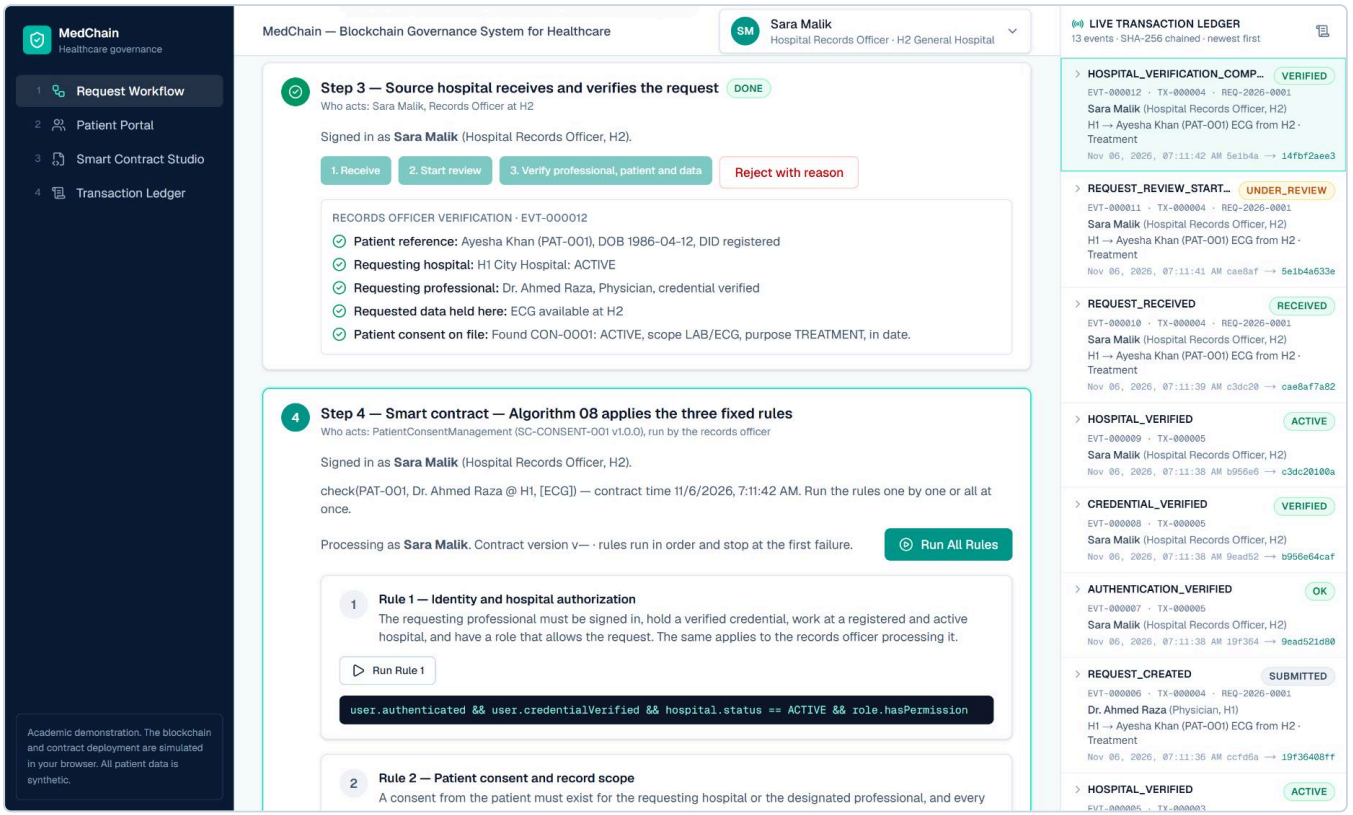


Figure 4. Step 3: receipt, review and verification recorded; verification checklist shown.

Live ledger: events written by this step (6)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000007 TX-000005	**AUTHENTICATION_VERIFIED** Sara Malik signed in as Hospital Records Officer using DID did:medchain:user:66ab9eea1309ceb0 (demo authentication).	Sara Malik Hospital Records Officer, H2	OK	19f36408ffbb… ↓ **9ead521d804b…**
EVT-000008 TX-000005	**CREDENTIAL_VERIFIED** Credential urn:vc:medchain:u-rec-h2 verified: trusted issuer, intact, in date, not revoked.	Sara Malik Hospital Records Officer, H2	VERIFIED	9ead521d804b… ↓ **b956e64cafb1…**
EVT-000009 TX-000005	**HOSPITAL_VERIFIED** H2 General Hospital is registered and active in the consortium registry.	Sara Malik Hospital Records Officer, H2	ACTIVE	b956e64cafb1… ↓ **c3dc20100a79…**
EVT-000010 TX-000004 REQ-2026-0001	**REQUEST_RECEIVED** H2 received REQ-2026-0001 in its inbox; Sara Malik acknowledged it.	Sara Malik Hospital Records Officer, H2	RECEIVED	c3dc20100a79… ↓ **cae8af7a82ca…**
EVT-000011 TX-000004 REQ-2026-0001	**REQUEST_REVIEW_STARTED** Sara Malik opened REQ-2026-0001 and started reviewing it.	Sara Malik Hospital Records Officer, H2	UNDER_REVIEW	cae8af7a82ca… ↓ **5e1b4a633ecc…**

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000012 TX-000004 REQ-2026-0001	**HOSPITAL_VERIFICATION_COMPLETED** Sara Malik verified REQ-2026-0001: patient reference OK, requesting hospital OK, requesting professional OK, requested data held here OK, patient consent on file OK. This is the hospital's check; Algorithm 08 still decides independently.	Sara Malik Hospital Records Officer, H2	VERIFIED	5e1b4a633ecc… ↓ **14fbf2aee34e…**

Step 4 — Smart contract — Algorithm 08 applies the three rules

> **Assignment requirement:** Algorithm 08 executes; all three rules are evaluated individually, each showing its actual inputs, PASS/FAIL result and explanation.

What the user does: Clicks *Run Rule 1*, *Run Rule 2* and *Run Rule 3* (or *Run All Rules*).

What the system checks (logic): R1: both professionals authenticated and verified, H1 and H2 active, roles permitted → PASS. R2: consent CON-0001 grants H1, scope LAB and ECG includes ECG → PASS. R3: purpose Treatment = Treatment, ACTIVE, before 31 Dec 2026 → PASS. Result: CONTRACT_APPROVED.

Why: Each rule is recorded as its own event with its inputs, so the decision can be audited later.

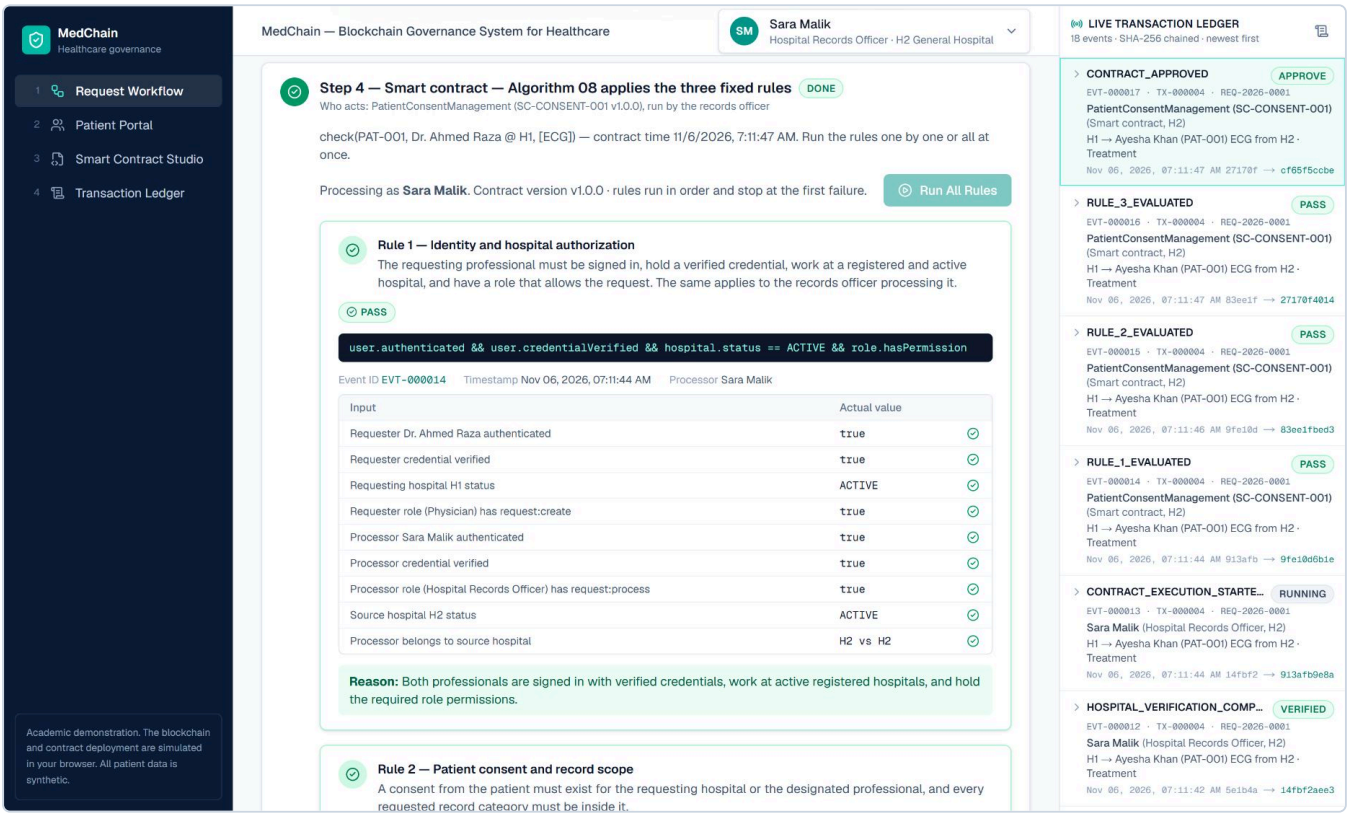


Figure 5. Rule 1 with every actual input and its result; the rule events appear in the live ledger.

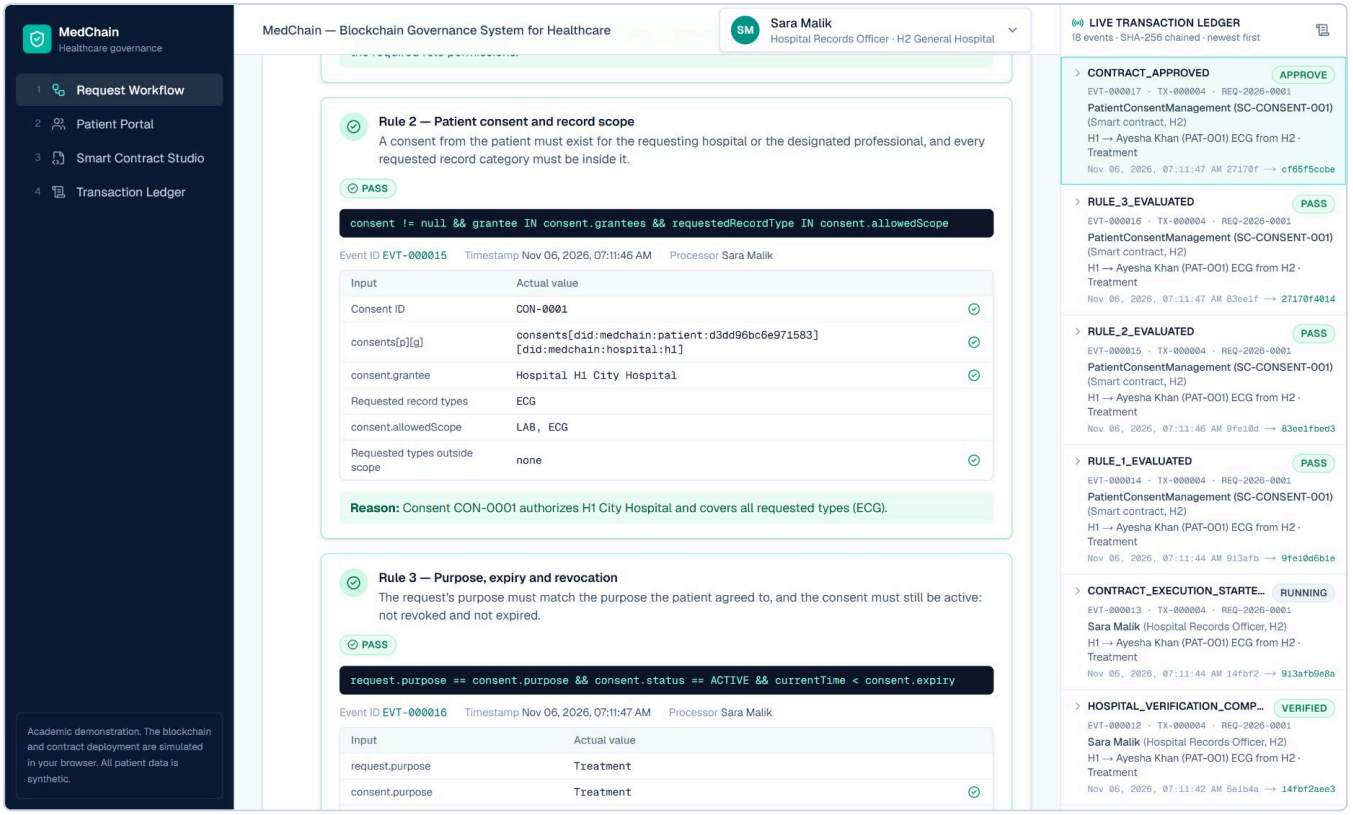


Figure 6. Rules 2 and 3 with their inputs; the contract approves (CONTRACT_APPROVED at the top of the live ledger).

Live ledger: events written by this step (5)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000013 TX-000004 REQ-2026-0001	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h1@H1, [ECG]).	Sara Malik Hospital Records Officer, H2	RUNNING	14fbf2aee34e… ↓ **913afb9e8a76…**
EVT-000014 TX-000004 REQ-2026-0001	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Both professionals are signed in with verified credentials, work at active registered hospitals, and hold the required role permissions.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	913afb9e8a76… ↓ **9fe10d6b1e7f…**
EVT-000015 TX-000004 REQ-2026-0001	**RULE_2_EVALUATED** Rule 2 (Patient consent and record scope): Consent CON-0001 authorizes H1 City Hospital and covers all requested types (ECG).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	9fe10d6b1e7f… ↓ **83ee1fbed31b…**
EVT-000016 TX-000004 REQ-2026-0001	**RULE_3_EVALUATED** Rule 3 (Purpose, expiry and revocation): Purpose matches and the consent is active, not revoked and not expired.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	83ee1fbed31b… ↓ **27170f4014e6…**
EVT-000017 TX-000004 REQ-2026-0001	**CONTRACT_APPROVED** check() returned APPROVE: all three fixed rules passed. The source hospital must now approve or reject.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	APPROVE	27170f4014e6… ↓ **cf65f5ccbebd…**

Step 5 — Source hospital responds

> **Assignment requirement:** The Records Officer approves or rejects with a reason; the hospital cannot override patient consent or the rules.

What the user does: Clicks *Approve request*. (A *Reject with reason* button is also available.)

What the system checks (logic): Approval is accepted only when the request's status is CONTRACT_APPROVED, so a denied request can never be approved.

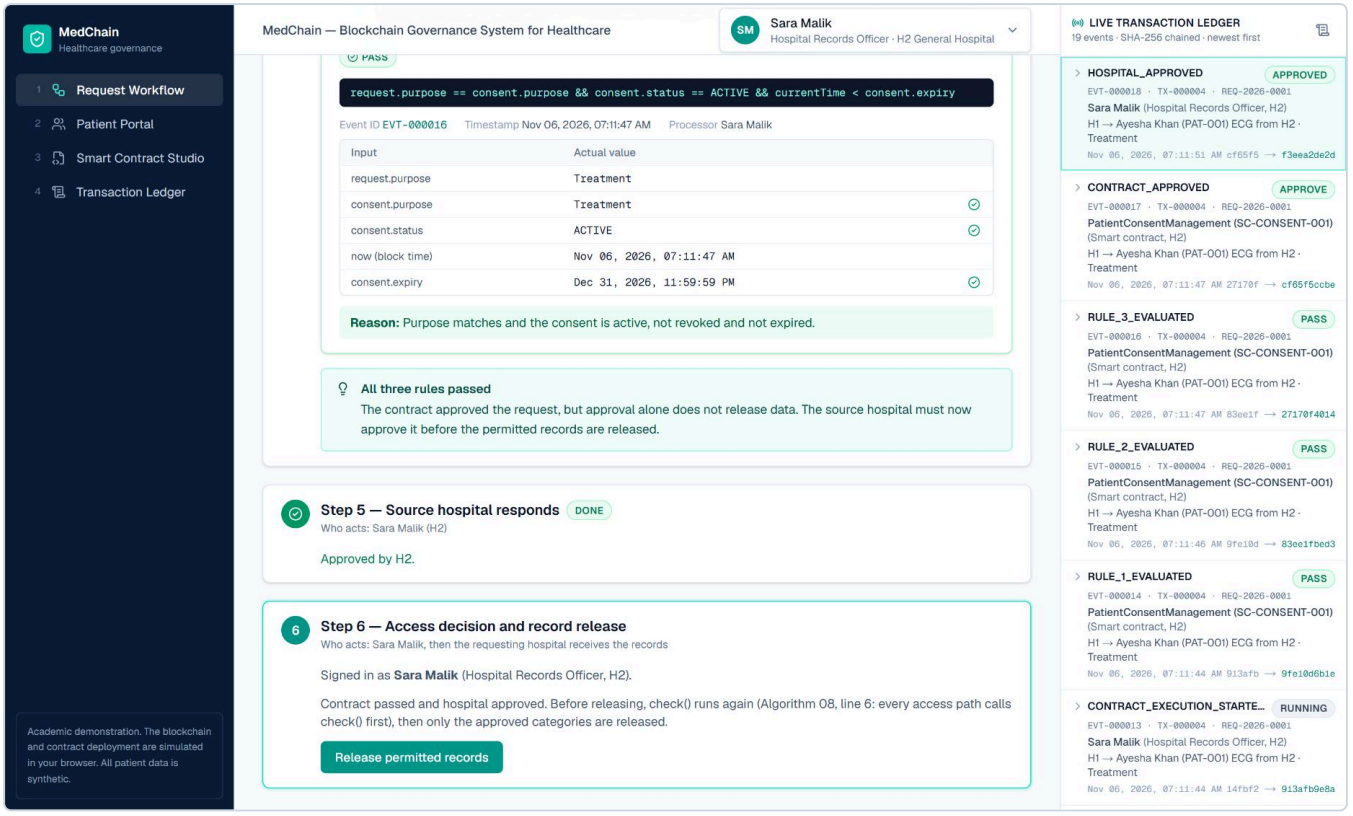


Figure 7. HOSPITAL_APPROVED recorded.

Live ledger: events written by this step (1)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000018 TX-000004 REQ-2026-0001	**HOSPITAL_APPROVED** Sara Malik (H2) approved REQ-2026-0001. Contract passed and hospital approved: the permitted records may now be released.	Sara Malik Hospital Records Officer, H2	APPROVED	cf65f5ccbebd... ↓ **f3eea2de2de5...**

Step 6 — Access decision and medical record release

> **Assignment requirement:** If the hospital approves and all rules pass, the permitted medical history is released; the final decision and reason are displayed.

What the user does: Clicks *Release permitted records*.

What the system checks (logic): Released records = records of this patient, held by the source hospital, whose category was requested **and** is inside the consent. Here only REC-001-ECG is released; LAB (consented but not requested) and MRI (not consented) are not.

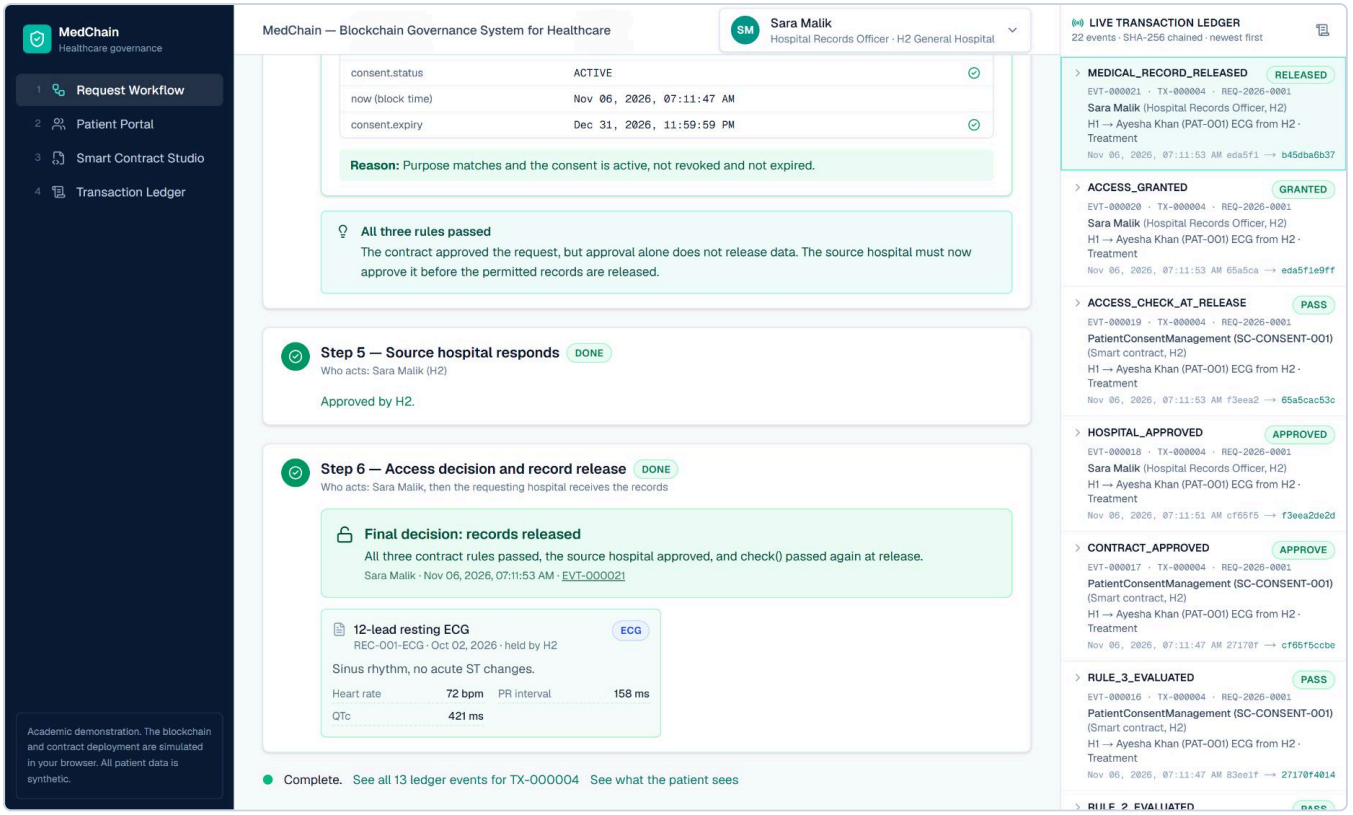


Figure 8. Final decision: records released. Only the ECG record is delivered.

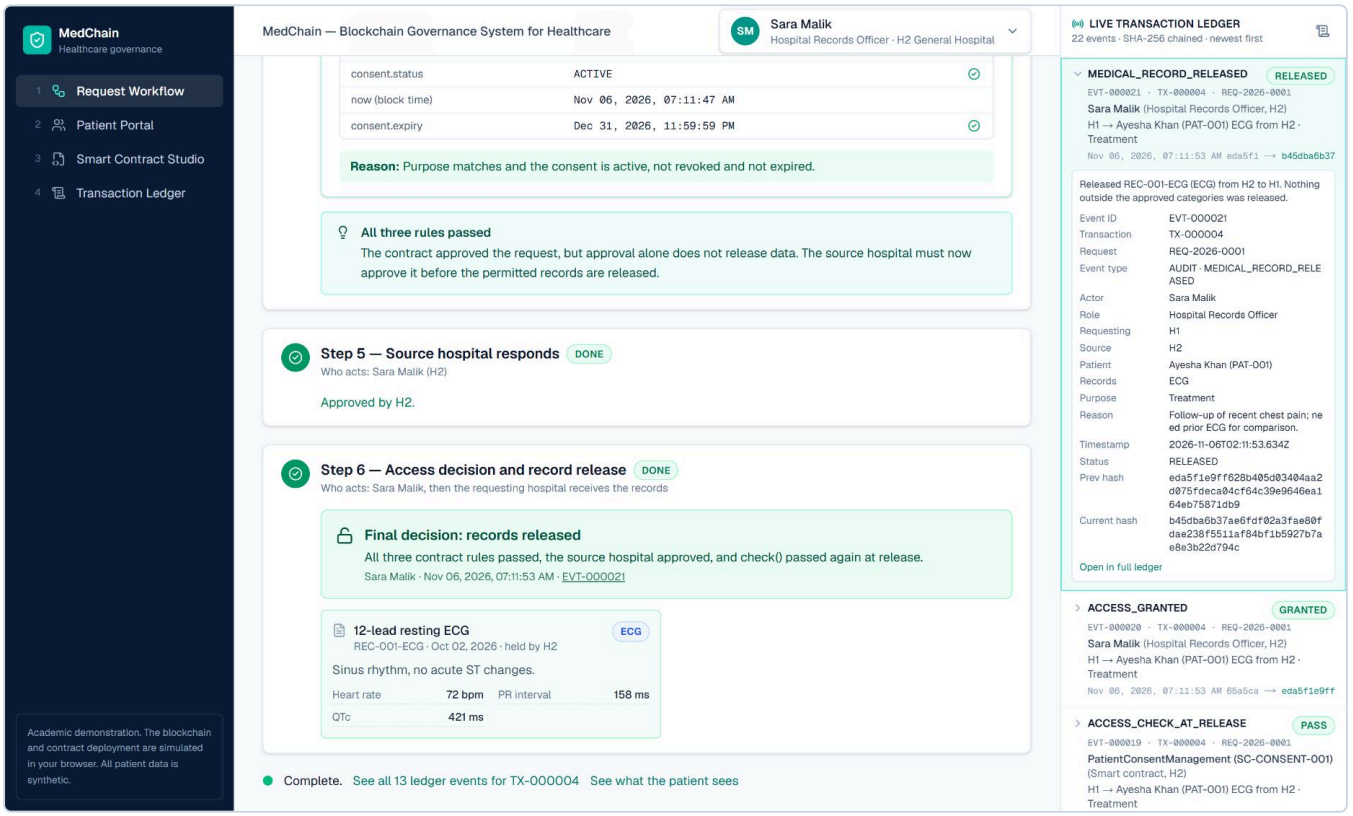


Figure 9. Live ledger with the MEDICAL_RECORD_RELEASED event expanded: event, transaction and request IDs, actor and role, requesting and source hospitals, patient, records, purpose, reason, timestamp and both full hashes.

Live ledger: events written by this step (3)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000019 TX-000004 REQ-2026-0001	**ACCESS_CHECK_AT_RELEASE** check() re-run at the moment of access: Rules 1–3 still pass, so the release may go ahead.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	f3eea2de2de5… ↓ **65a5cac53ca1…**
EVT-000020 TX-000004 REQ-2026-0001	**ACCESS_GRANTED** Access granted: all three contract rules passed and the source hospital approved.	Sara Malik Hospital Records Officer, H2	GRANTED	65a5cac53ca1… ↓ **eda5f1e9ff62…**
EVT-000021 TX-000004 REQ-2026-0001	**MEDICAL_RECORD_RELEASED** Released REC-001-ECG (ECG) from H2 to H1. Nothing outside the approved categories was released.	Sara Malik Hospital Records Officer, H2	RELEASED	eda5f1e9ff62… ↓ **b45dba6b37ae…**

7. Case 2 — MRI request outside consent (denied by Rule 2)

Same physician and patient, but the request asks for **MRI**, which the consent does not include. Request **REQ-2026-0002**, transaction **TX-000007**.

Request submitted

What the user does: Starts a new request, signs in as Dr. Ahmed Raza again, ticks MRI only, submits.

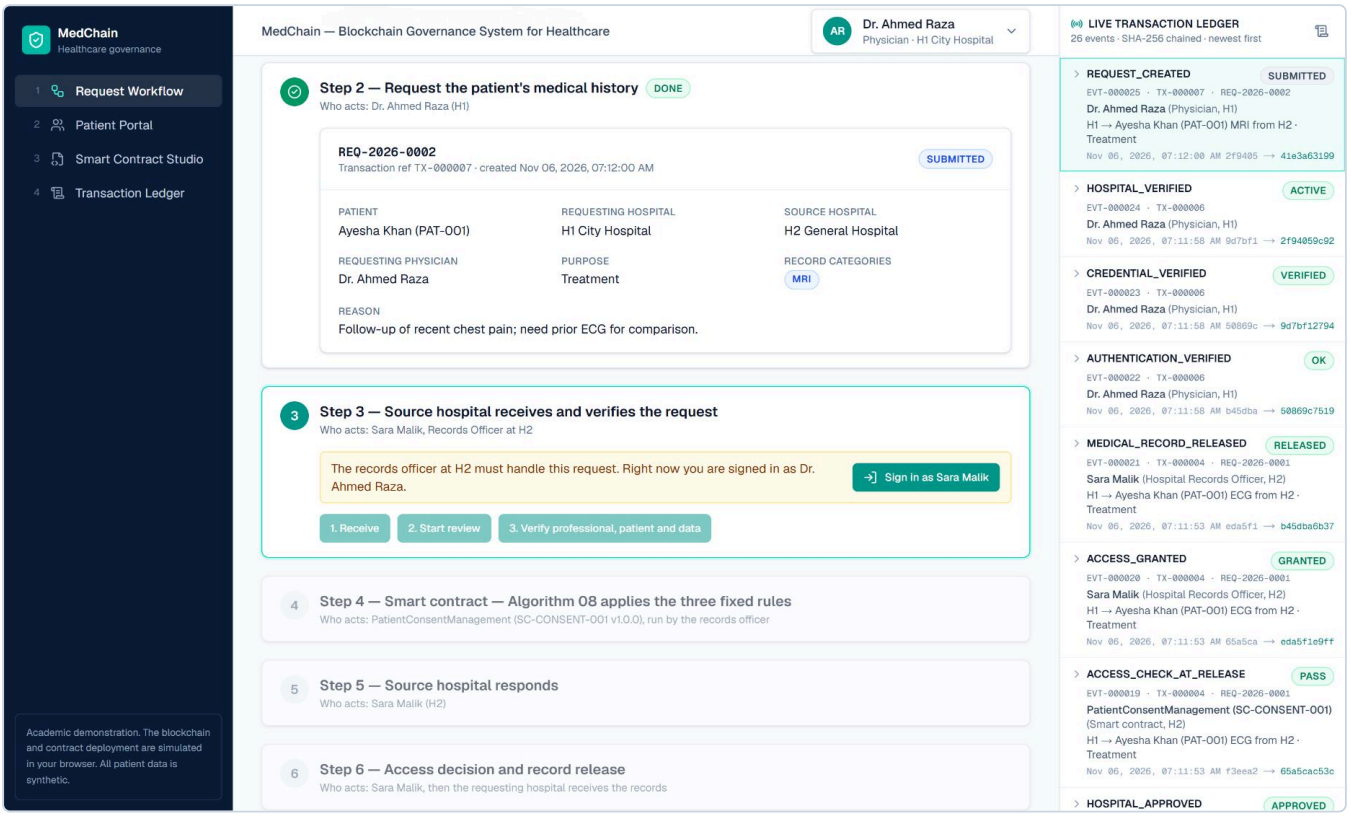


Figure 10. MRI request submitted; REQUEST_CREATED at the top of the live ledger.

Live ledger: events written by this step (4)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000022 TX-000006	**AUTHENTICATION_VERIFIED** Dr. Ahmed Raza signed in as Physician using DID did:medchain:user:bdbf7bc7b1f566b7 (demo authentication).	Dr. Ahmed Raza Physician, H1	OK	b45dba6b37ae… ↓ **50869c7519f8…**
EVT-000023 TX-000006	**CREDENTIAL_VERIFIED** Credential urn:vc:medchain:u-phy-h1 verified: trusted issuer, intact, in date, not revoked.	Dr. Ahmed Raza Physician, H1	VERIFIED	50869c7519f8… ↓ **9d7bf12794fc…**
EVT-000024 TX-000006	**HOSPITAL_VERIFIED** H1 City Hospital is registered and active in the consortium registry.	Dr. Ahmed Raza Physician, H1	ACTIVE	9d7bf12794fc… ↓ **2f94059c9202…**
EVT-000025 TX-000007 REQ-2026-0002	**REQUEST_CREATED** Dr. Ahmed Raza (H1) requested Ayesha Khan's MRI from H2 for Treatment. Reason: Follow-up of recent chest pain; need prior ECG for comparison.	Dr. Ahmed Raza Physician, H1	SUBMITTED	2f94059c9202… ↓ **41e3a63199ff…**

Hospital verification and contract execution

What the system checks (logic): R1 passes (same authorized people). R2: requested MRI is not in the allowed scope (LAB, ECG) → FAIL. check() stops; R3 is not evaluated. CONTRACT_DENIED and ACCESS_DENIED are

recorded.

Why: The patient never consented to MRI sharing, so no hospital decision can release it.

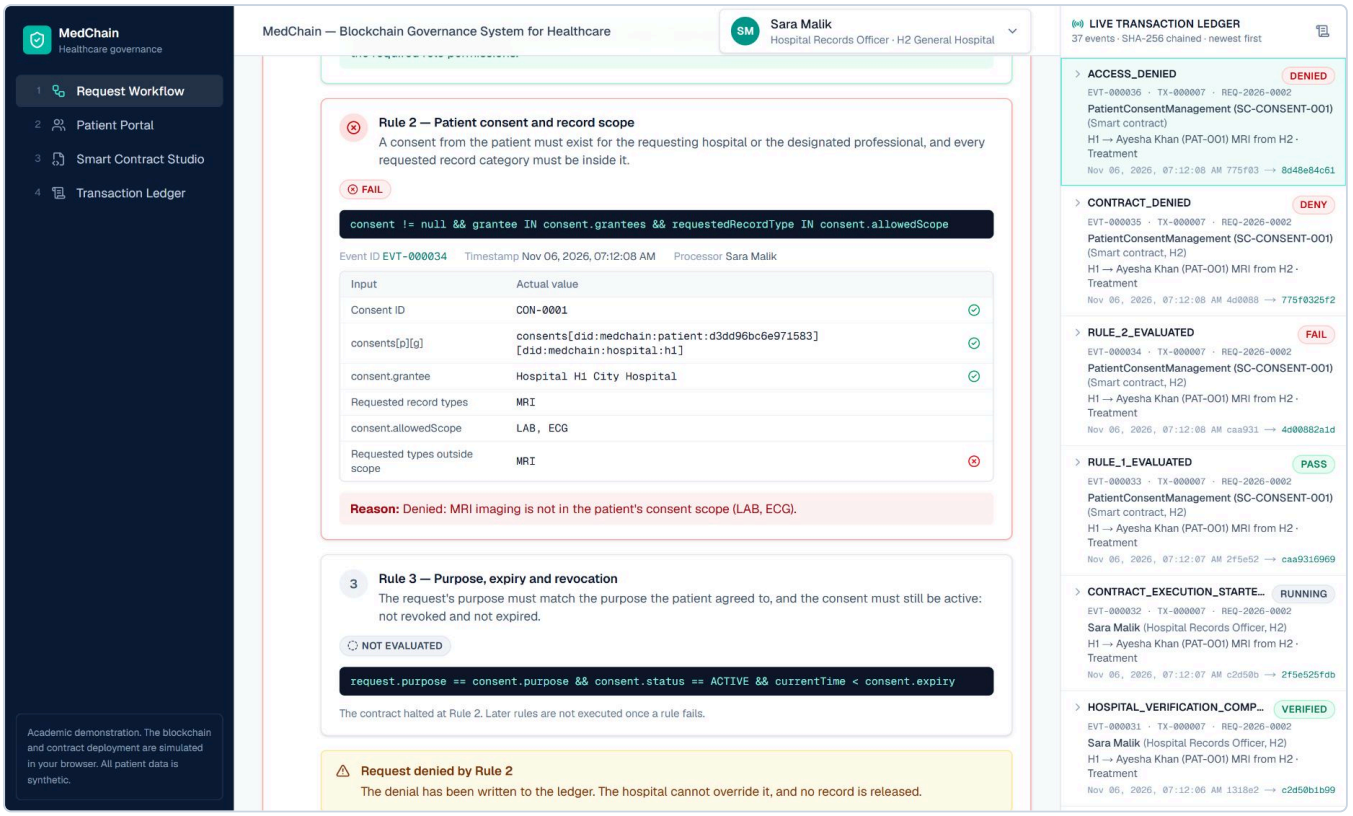


Figure 11. Rule 2 FAIL with its inputs: requested MRI, allowed LAB and ECG.

Live ledger: events written by this step (11)

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000026 TX-000008	**AUTHENTICATION_VERIFIED** Sara Malik signed in as Hospital Records Officer using DID did:medchain:user:66ab9eea1309ceb0 (demo authentication).	Sara Malik Hospital Records Officer, H2	OK	41e3a63199ff… ↓ **ca5a6389d6ad…**
EVT-000027 TX-000008	**CREDENTIAL_VERIFIED** Credential urn:vc:medchain:u-rec-h2 verified: trusted issuer, intact, in date, not revoked.	Sara Malik Hospital Records Officer, H2	VERIFIED	ca5a6389d6ad… ↓ **f2cd2b7a2477…**
EVT-000028 TX-000008	**HOSPITAL_VERIFIED** H2 General Hospital is registered and active in the consortium registry.	Sara Malik Hospital Records Officer, H2	ACTIVE	f2cd2b7a2477… ↓ **09daa105bfaa…**
EVT-000029 TX-000007 REQ-2026-0002	**REQUEST_RECEIVED** H2 received REQ-2026-0002 in its inbox; Sara Malik acknowledged it.	Sara Malik Hospital Records Officer, H2	RECEIVED	09daa105bfaa… ↓ **4b3ea76eaf84…**
EVT-000030 TX-000007 REQ-2026-0002	**REQUEST_REVIEW_STARTED** Sara Malik opened REQ-2026-0002 and started reviewing it.	Sara Malik Hospital Records Officer, H2	UNDER_REVIEW	4b3ea76eaf84… ↓ **1318e21715aa…**
EVT-000031 TX-000007 REQ-2026-0002	**HOSPITAL_VERIFICATION_COMPLETED** Sara Malik verified REQ-2026-0002: patient reference OK, requesting hospital OK, requesting professional OK, requested data held here OK, patient consent on file OK. This is the hospital's check; Algorithm 08 still decides independently.	Sara Malik Hospital Records Officer, H2	VERIFIED	1318e21715aa… ↓ **c2d50b1b9901…**

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000032 TX-000007 REQ-2026-0002	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h1@H1, [MRI]).	Sara Malik Hospital Records Officer, H2	RUNNING	c2d50b1b9901… ↓ **2f5e525fdb2b…**
EVT-000033 TX-000007 REQ-2026-0002	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Both professionals are signed in with verified credentials, work at active registered hospitals, and hold the required role permissions.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	2f5e525fdb2b… ↓ **caa9316969fb…**
EVT-000034 TX-000007 REQ-2026-0002	**RULE_2_EVALUATED** Rule 2 (Patient consent and record scope): Denied: MRI imaging is not in the patient's consent scope (LAB, ECG).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	FAIL	caa9316969fb… ↓ **4d00882a1d3f…**
EVT-000035 TX-000007 REQ-2026-0002	**CONTRACT_DENIED** check() returned DENY at Rule 2. A failed rule cannot be overridden.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	DENY	4d00882a1d3f… ↓ **775f0325f2c6…**
EVT-000036 TX-000007 REQ-2026-0002	**ACCESS_DENIED** Access denied. Denied: MRI imaging is not in the patient's consent scope (LAB, ECG). No medical records were released.	PatientConsentManagement (SC-CONSENT-001) Smart contract	DENIED	775f0325f2c6… ↓ **8d48e84c6140…**

Outcome

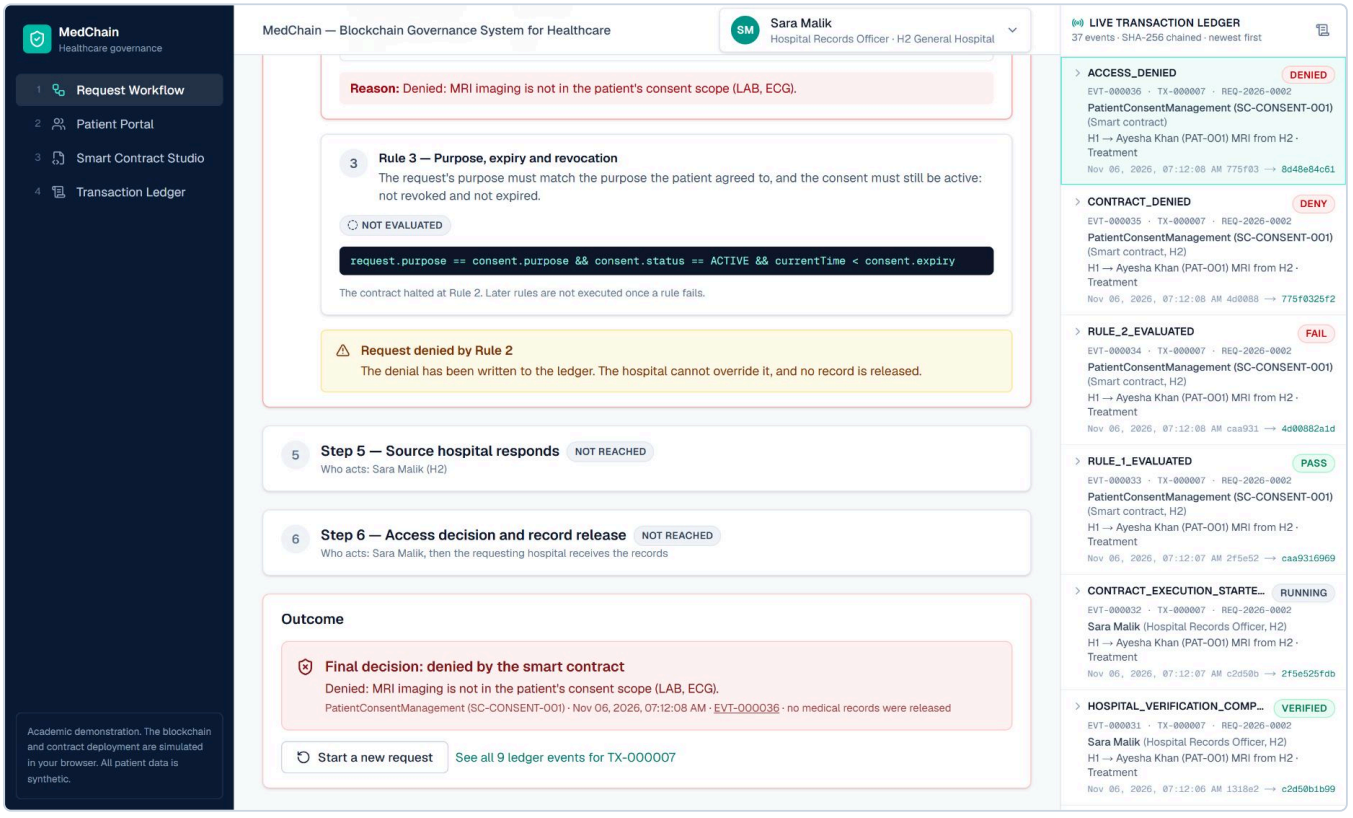


Figure 12. Steps 5 and 6 are 'Not reached'. No Approve or Release action exists. Final decision: denied by the smart contract; no medical records released.

8. Further denial cases

8.1 Unauthorized hospital — denied by Rule 1 (REQ-2026-0003)

Dr. Victor Hale works at H5 External Hospital, which is not registered. His credential is issued by H5, which is not a trusted consortium issuer.

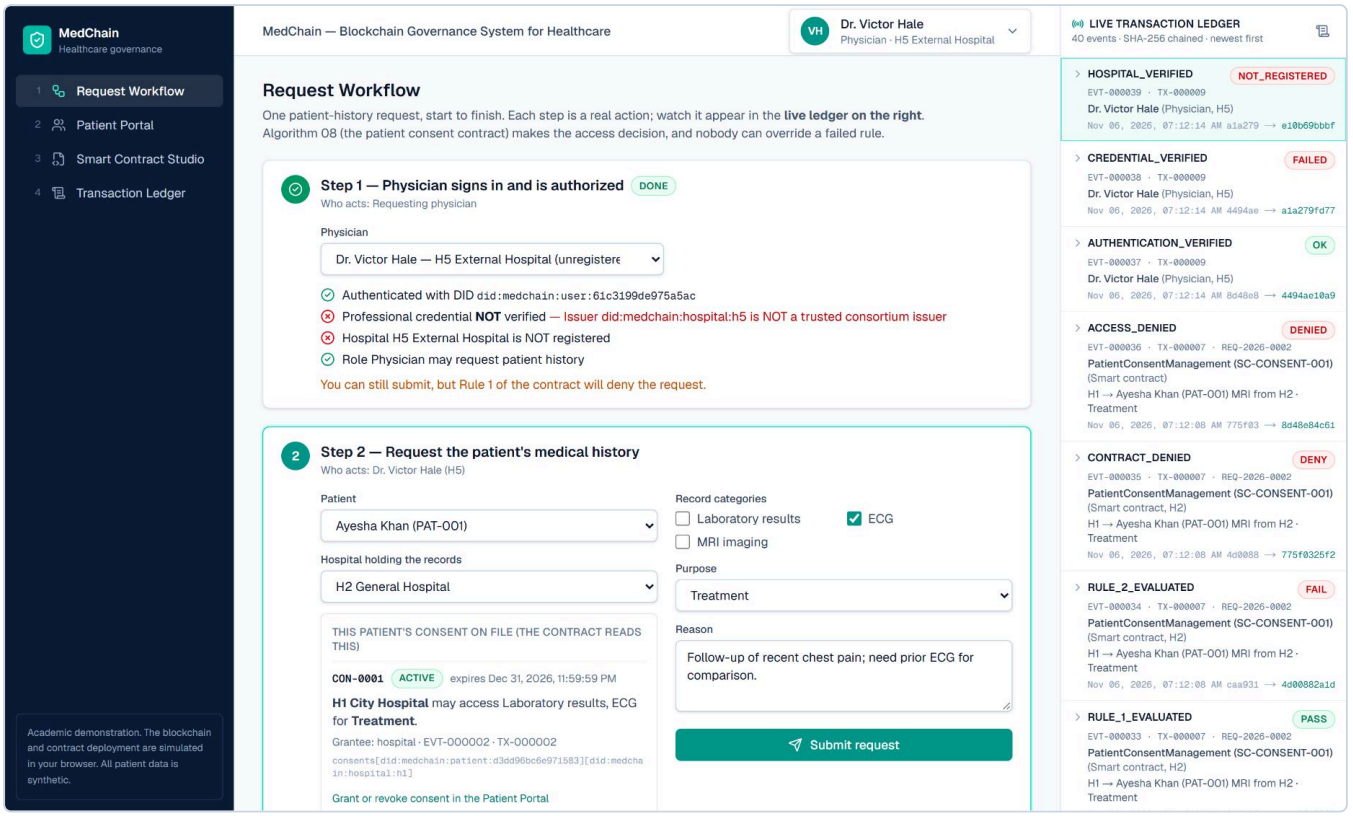


Figure 13. Sign-in: authenticated, but credential verification fails and the hospital is not registered. Submitting is allowed so that the contract makes the decision.

Events written at sign-in

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000037 TX-000009	**AUTHENTICATION_VERIFIED** Dr. Victor Hale signed in as Physician using DID did:medchain:user:61c3199de975a5ac (demo authentication).	Dr. Victor Hale Physician, H5	OK	8d48e84c6140… ↓ **4494ae10a94e…**
EVT-000038 TX-000009	**CREDENTIAL_VERIFIED** Credential verification failed: Issuer did:medchain:hospital:h5 is NOT a trusted consortium issuer.	Dr. Victor Hale Physician, H5	FAILED	4494ae10a94e… ↓ **a1a279fd7735…**
EVT-000039 TX-000009	**HOSPITAL_VERIFIED** H5 External Hospital is not registered in the consortium registry.	Dr. Victor Hale Physician, H5	NOT_REGISTERED	a1a279fd7735… ↓ **e10b69bbbfb9…**

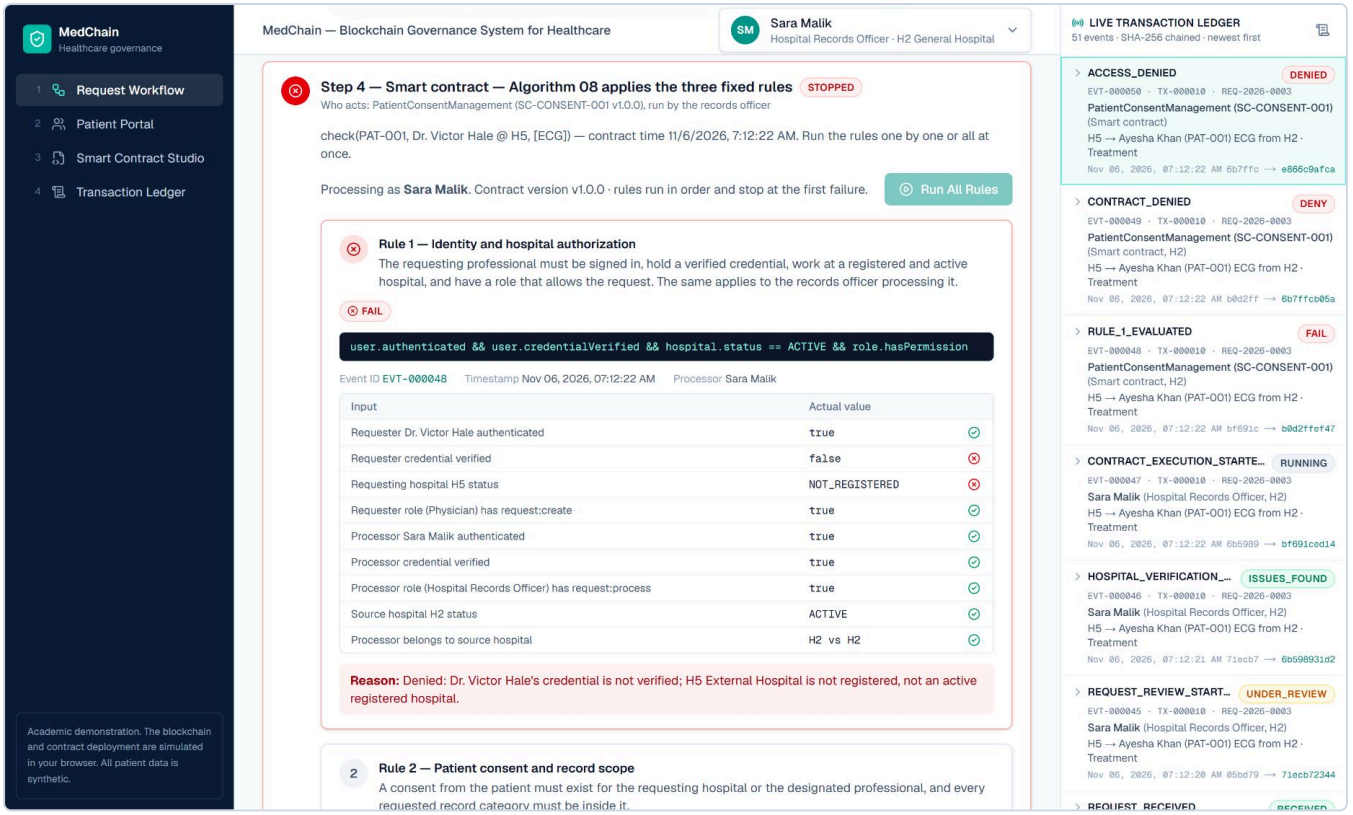


Figure 14. Rule 1 FAIL: requester credential not verified and H5 not ACTIVE. Rules 2 and 3 are not evaluated.

Events of this request

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000040 TX-000010 REQ-2026-0003	**REQUEST_CREATED** Dr. Victor Hale (H5) requested Ayesha Khan's ECG from H2 for Treatment. Reason: Follow-up of recent chest pain; need prior ECG for comparison.	Dr. Victor Hale Physician, H5	SUBMITTED	e10b69bbbfb9… ↓ **7ed54f18e1cf…**
EVT-000044 TX-000010 REQ-2026-0003	**REQUEST_RECEIVED** H2 received REQ-2026-0003 in its inbox; Sara Malik acknowledged it.	Sara Malik Hospital Records Officer, H2	RECEIVED	7a89e3f80693… ↓ **05bd7978af53…**
EVT-000045 TX-000010 REQ-2026-0003	**REQUEST_REVIEW_STARTED** Sara Malik opened REQ-2026-0003 and started reviewing it.	Sara Malik Hospital Records Officer, H2	UNDER_REVIEW	05bd7978af53… ↓ **71ecb723443a…**
EVT-000046 TX-000010 REQ-2026-0003	**HOSPITAL_VERIFICATION_COMPLETED** Sara Malik verified REQ-2026-0003: patient reference OK, requesting hospital flagged, requesting professional flagged, requested data held here OK, patient consent on file flagged. This is the hospital's check; Algorithm 08 still decides independently.	Sara Malik Hospital Records Officer, H2	ISSUES_FOUND	71ecb723443a… ↓ **6b598931d2e8…**
EVT-000047 TX-000010 REQ-2026-0003	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h5@H5, [ECG]).	Sara Malik Hospital Records Officer, H2	RUNNING	6b598931d2e8… ↓ **bf691ced144f…**
EVT-000048 TX-000010 REQ-2026-0003	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Denied: Dr. Victor Hale's credential is not verified; H5 External Hospital is not registered, not an active registered hospital.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	FAIL	bf691ced144f… ↓ **b0d2ffef4786…**
EVT-000049 TX-000010 REQ-2026-0003	**CONTRACT_DENIED** check() returned DENY at Rule 1. A failed rule cannot be overridden.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	DENY	b0d2ffef4786… ↓ **6b7ffcb05a12…**

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000050 TX-000010 REQ-2026-0003	**ACCESS_DENIED** Access denied. Denied: Dr. Victor Hale's credential is not verified; H5 External Hospital is not registered, not an active registered hospital. No medical records were released.	PatientConsentManagement (SC-CONSENT-001) Smart contract	DENIED	6b7ffcb05a12… ↓ **e866c9afca79…**

8.2 Revoked consent — denied by Rule 3 (REQ-2026-0004)

Ayesha revokes City Hospital's permission in the Patient Portal (`revoke(PAT-001, H1)`). Dr. Ahmed Raza then requests the ECG again.

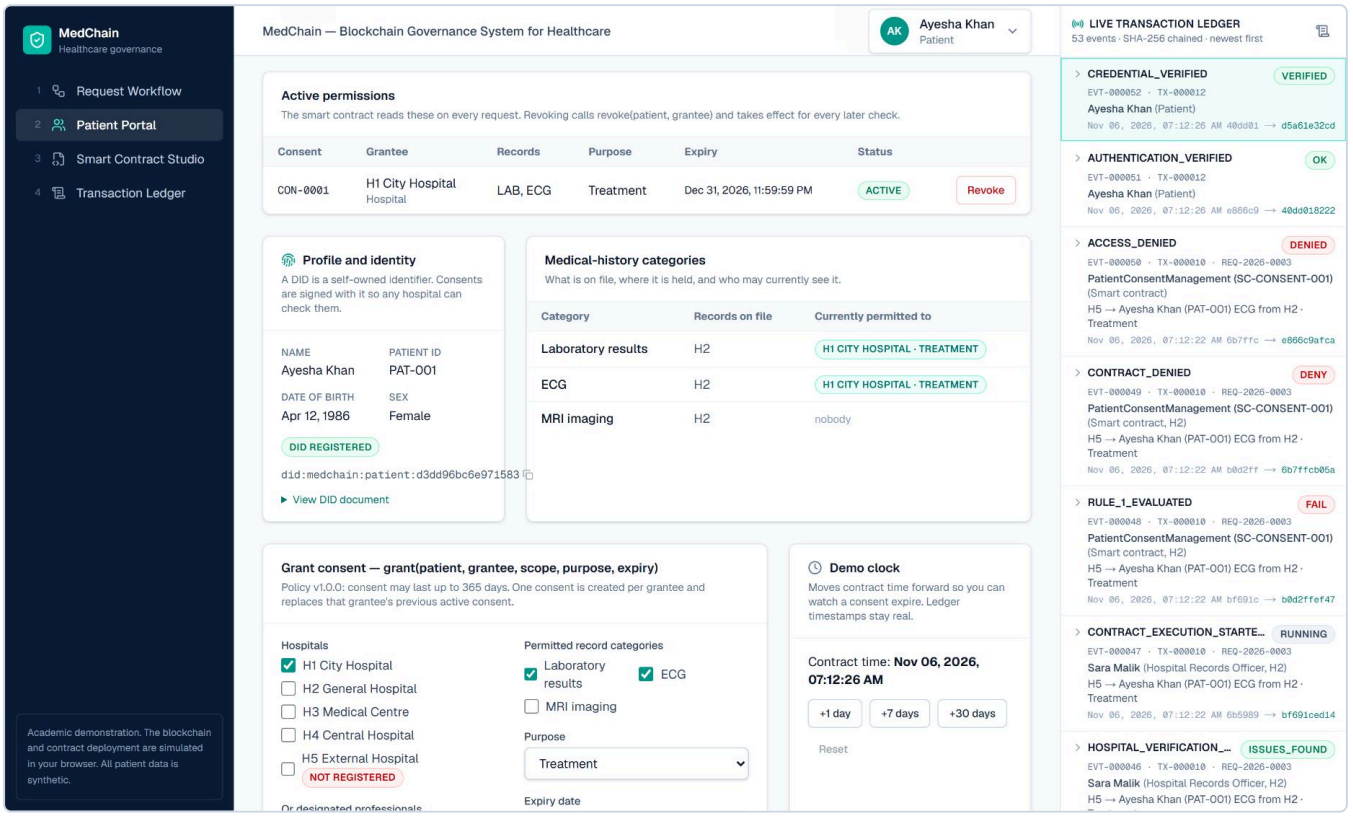


Figure 15. Before: City Hospital may access LAB and ECG for Treatment until 31 Dec 2026.

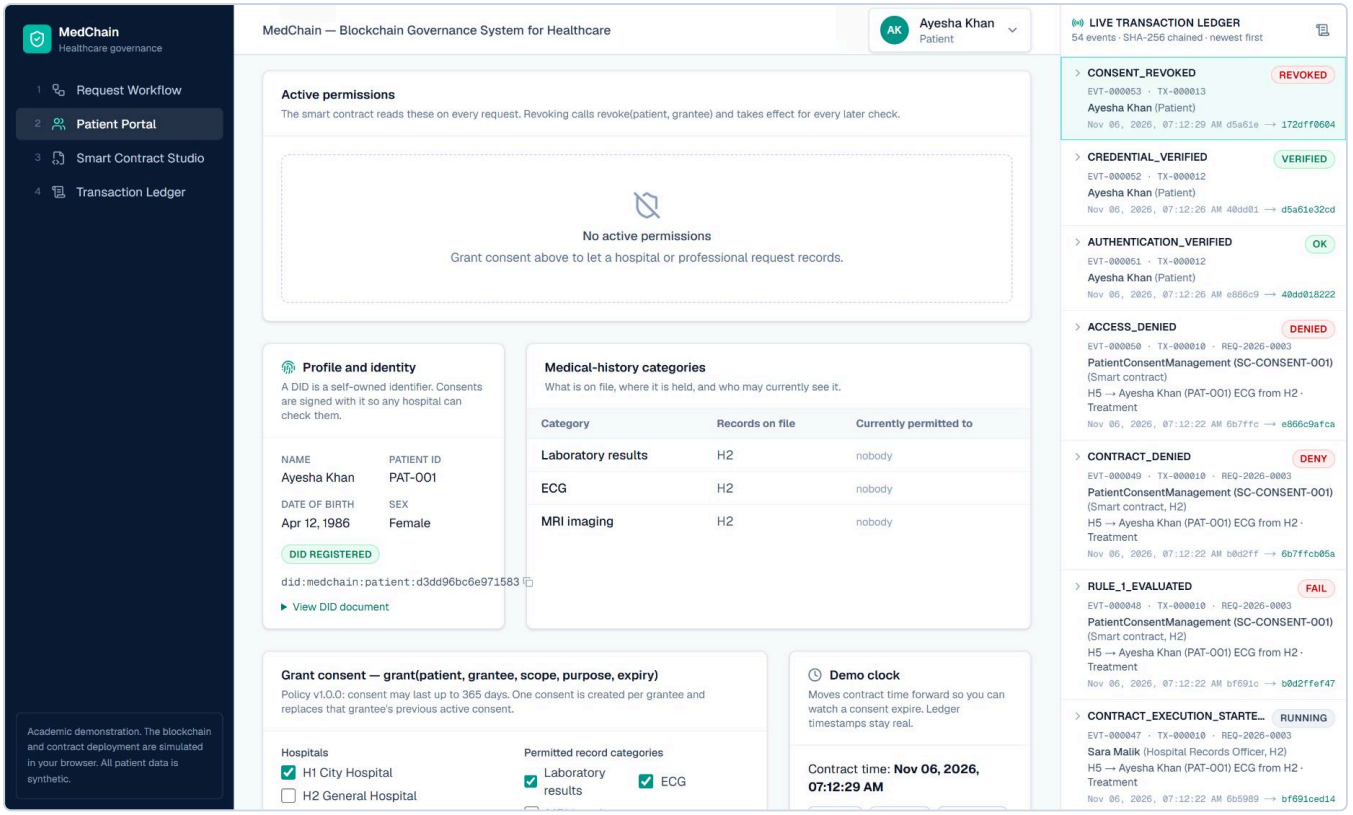


Figure 16. After revoke(): no active permissions; CONSENT_REVOKED at the top of the live ledger.

Revocation event

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000053 TX-000013	**CONSENT_REVOKED** revoke(did:medchain:patient:d3dd96bc6e971583, did:medchain:hospital:h1): Ayesha Khan's permission for H1 City Hospital withdrawn. Every later check() for this pair fails Rule 3.	Ayesha Khan Patient	REVOKED	d5a61e32cde4… ↓ **172dff06049f…**

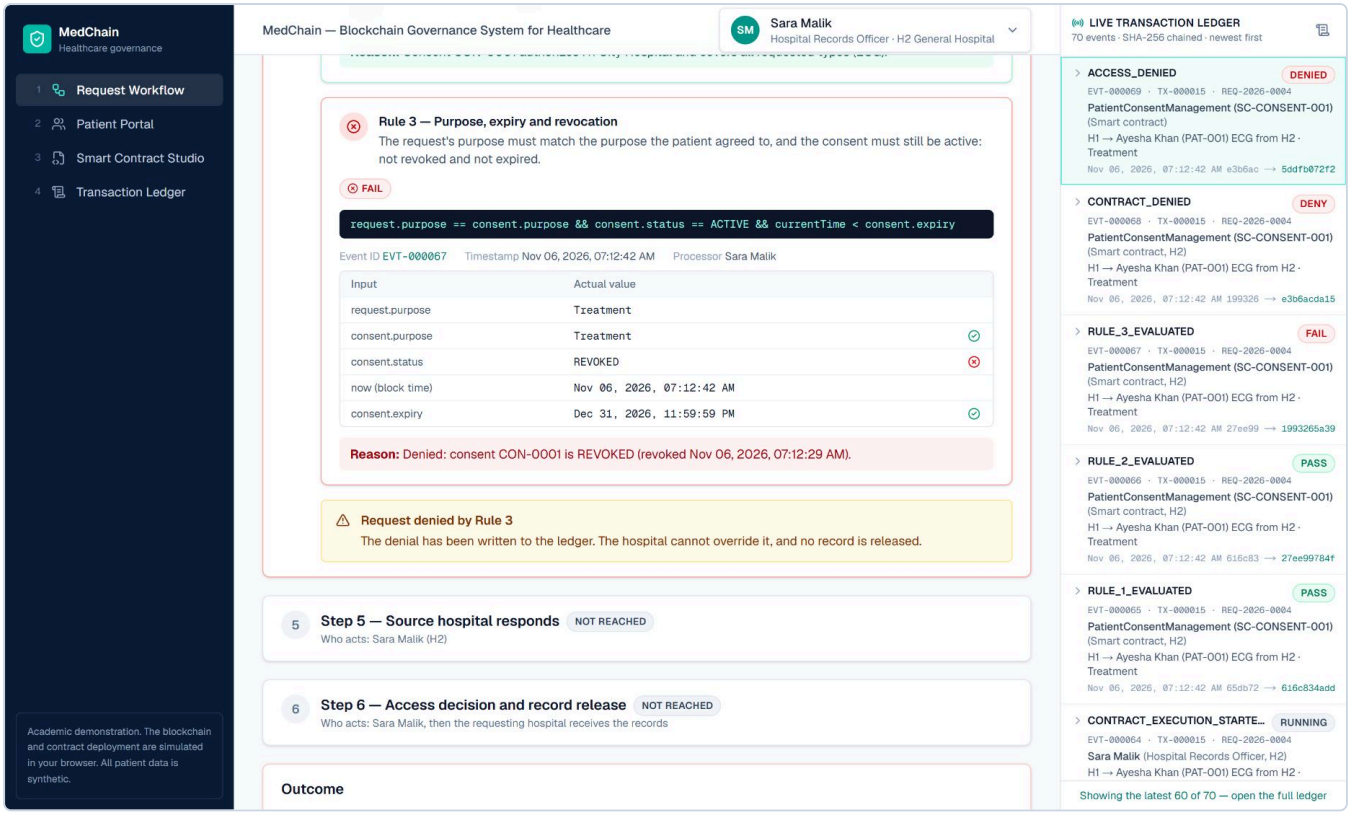


Figure 17. R1 and R2 pass (the revoked consent still names H1 and ECG), R3 fails: consent.status = REVOKED.

Contract and access events of this request

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000064 TX-000015 REQ-2026-0004	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h1@H1, [ECG]).	Sara Malik Hospital Records Officer, H2	RUNNING	5645ebf38b85… ↓ **65db723f52f4…**
EVT-000065 TX-000015 REQ-2026-0004	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Both professionals are signed in with verified credentials, work at active registered hospitals, and hold the required role permissions.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	65db723f52f4… ↓ **616c834add6b…**
EVT-000066 TX-000015 REQ-2026-0004	**RULE_2_EVALUATED** Rule 2 (Patient consent and record scope): Consent CON-0001 authorizes H1 City Hospital and covers all requested types (ECG).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	616c834add6b… ↓ **27ee99784fe8…**
EVT-000067 TX-000015 REQ-2026-0004	**RULE_3_EVALUATED** Rule 3 (Purpose, expiry and revocation): Denied: consent CON-0001 is REVOKED (revoked Nov 06, 2026, 07:12:29 AM).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	FAIL	27ee99784fe8… ↓ **1993265a39ff…**
EVT-000068 TX-000015 REQ-2026-0004	**CONTRACT_DENIED** check() returned DENY at Rule 3. A failed rule cannot be overridden.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	DENY	1993265a39ff… ↓ **e3b6acda1554…**
EVT-000069 TX-000015 REQ-2026-0004	**ACCESS_DENIED** Access denied. Denied: consent CON-0001 is REVOKED (revoked Nov 06, 2026, 07:12:29 AM). No medical records were released.	PatientConsentManagement (SC-CONSENT-001) Smart contract	DENIED	e3b6acda1554… ↓ **5ddfb072f237…**

Reasoning: revocation takes effect on the very next check because check() reads the current consent status at execution time; nothing is cached.

8.3 Expired consent — denied by Rule 3 (REQ-2026-0005)

Ayesha grants a new 7-day consent to City Hospital. The demo clock (used only for contract time) is then moved 14 days ahead to simulate the consent running out. Ledger timestamps stay real.

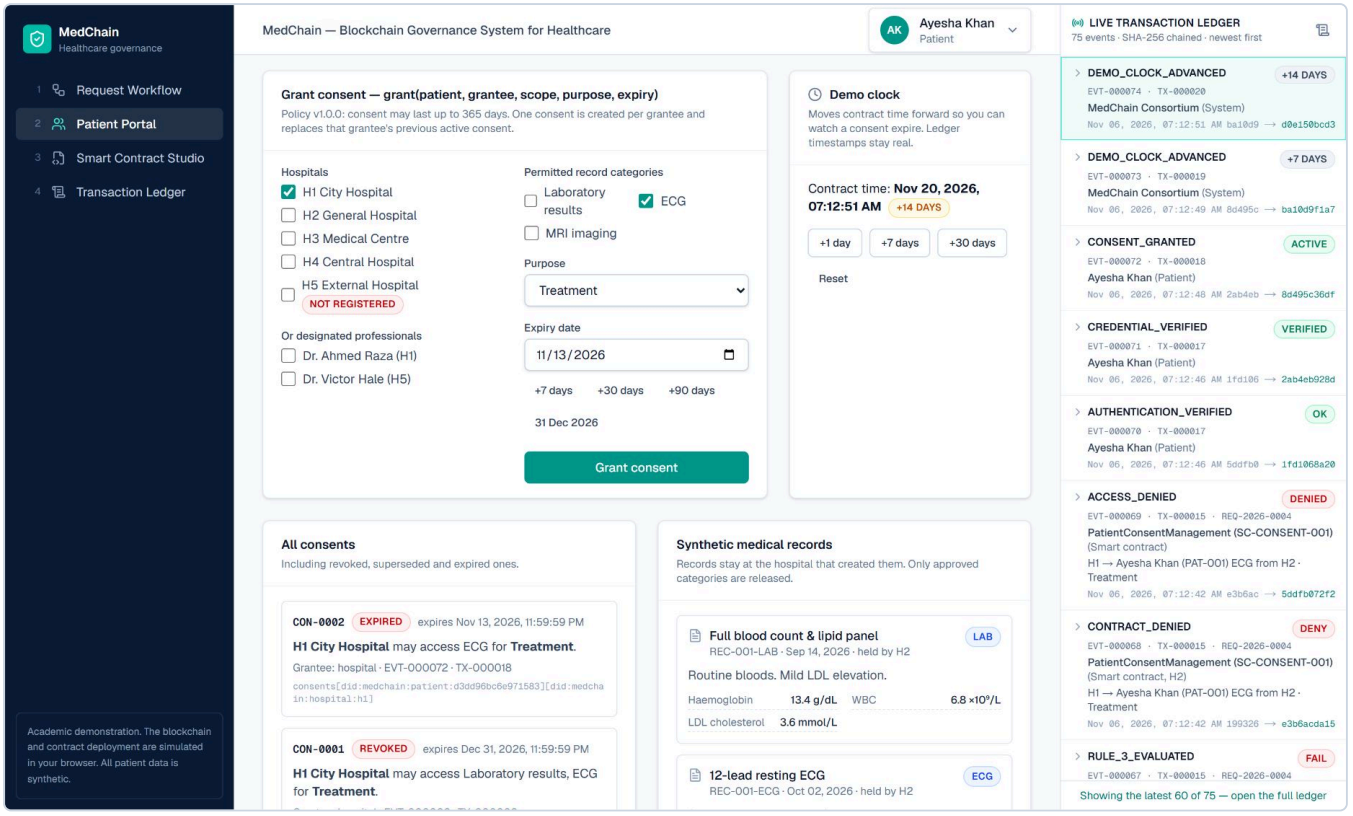


Figure 18. New 7-day consent granted; the demo clock is advanced. Each action is recorded in the live ledger.

Consent and clock events

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000072 TX-000018	**CONSENT_GRANTED** grant(did:medchain:patient:d3dd96bc6e971583, did:medchain:hospital:h1, [ECG], TREATMENT, 2026-11-13): Ayesha Khan allowed H1 City Hospital to access ECG for Treatment. Stored as consents[p][g].	Ayesha Khan Patient	ACTIVE	2ab4eb928dc8… ↓ **8d495c36dff9…**
EVT-000073 TX-000019	**DEMO_CLOCK_ADVANCED** Demo clock moved forward 7 day(s). Contract time is now Nov 13, 2026, 07:12:49 AM. Event timestamps stay real.	MedChain Consortium System	+7 days	8d495c36dff9… ↓ **ba10d9f1a7e8…**
EVT-000074 TX-000020	**DEMO_CLOCK_ADVANCED** Demo clock moved forward 7 day(s). Contract time is now Nov 20, 2026, 07:12:51 AM. Event timestamps stay real.	MedChain Consortium System	+14 days	ba10d9f1a7e8… ↓ **d0e150bcd39e…**

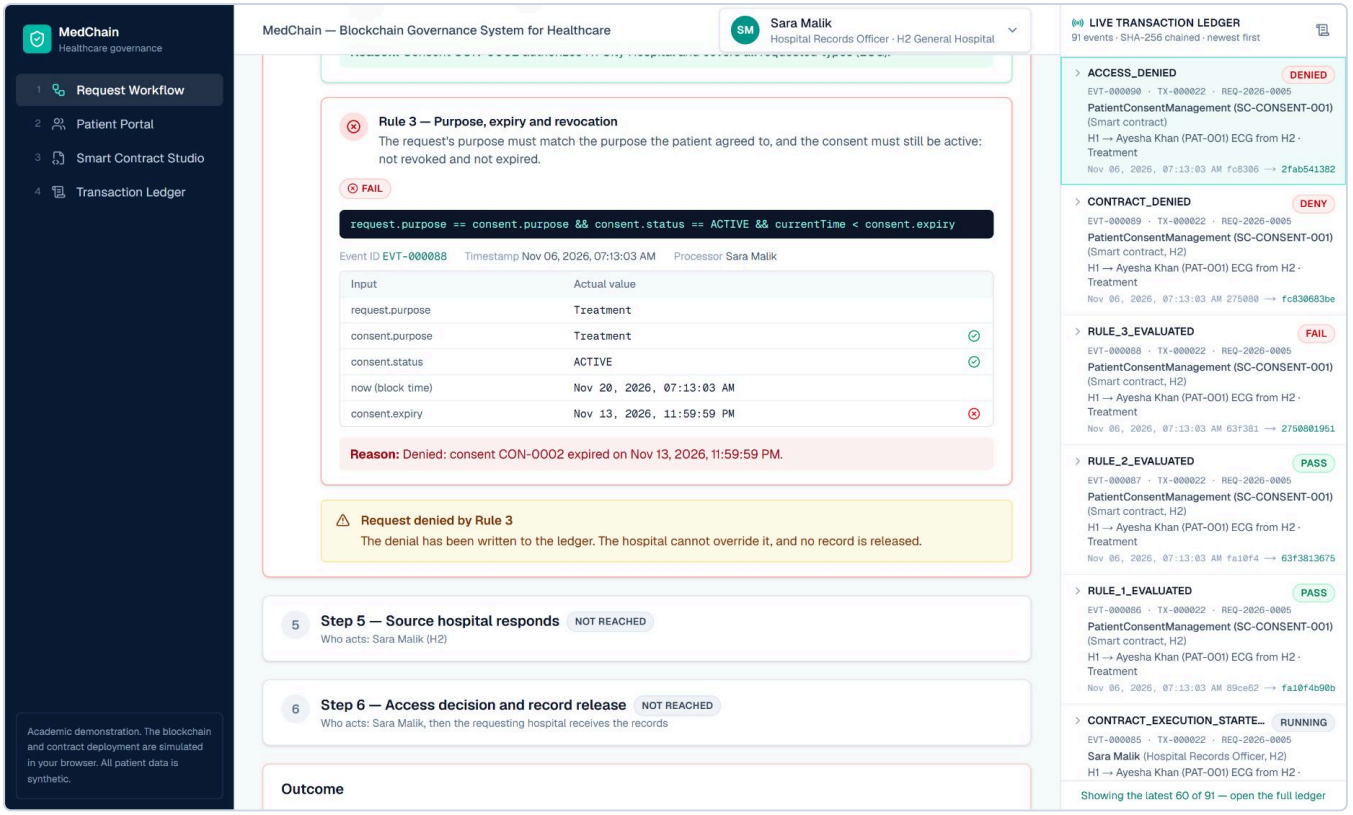


Figure 19. R3 FAIL: currentTime is after consent.expiry.

Contract and access events of this request

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000085 TX-000022 REQ-2026-0005	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h1@H1, [ECG]).	Sara Malik Hospital Records Officer, H2	RUNNING	6145acd33b25… ↓ **89ce626d8f0f…**
EVT-000086 TX-000022 REQ-2026-0005	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Both professionals are signed in with verified credentials, work at active registered hospitals, and hold the required role permissions.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	89ce626d8f0f… ↓ **fa10f4b90b36…**
EVT-000087 TX-000022 REQ-2026-0005	**RULE_2_EVALUATED** Rule 2 (Patient consent and record scope): Consent CON-0002 authorizes H1 City Hospital and covers all requested types (ECG).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	PASS	fa10f4b90b36… ↓ **63f381367516…**
EVT-000088 TX-000022 REQ-2026-0005	**RULE_3_EVALUATED** Rule 3 (Purpose, expiry and revocation): Denied: consent CON-0002 expired on Nov 13, 2026, 11:59:59 PM.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	FAIL	63f381367516… ↓ **2750801951f1…**
EVT-000089 TX-000022 REQ-2026-0005	**CONTRACT_DENIED** check() returned DENY at Rule 3. A failed rule cannot be overridden.	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	DENY	2750801951f1… ↓ **fc830683be35…**
EVT-000090 TX-000022 REQ-2026-0005	**ACCESS_DENIED** Access denied. Denied: consent CON-0002 expired on Nov 13, 2026, 11:59:59 PM. No medical records were released.	PatientConsentManagement (SC-CONSENT-001) Smart contract	DENIED	fc830683be35… ↓ **2fab54138215…**

8.4 Consent revoked after approval — blocked at release (REQ-2026-0006)

This demonstrates line 6 of Algorithm 08 (every access path calls check() first). The contract approves the request and the hospital approves it. Before the records are released, Ayesha revokes City Hospital's permission. When the officer clicks Release, check() is run again.

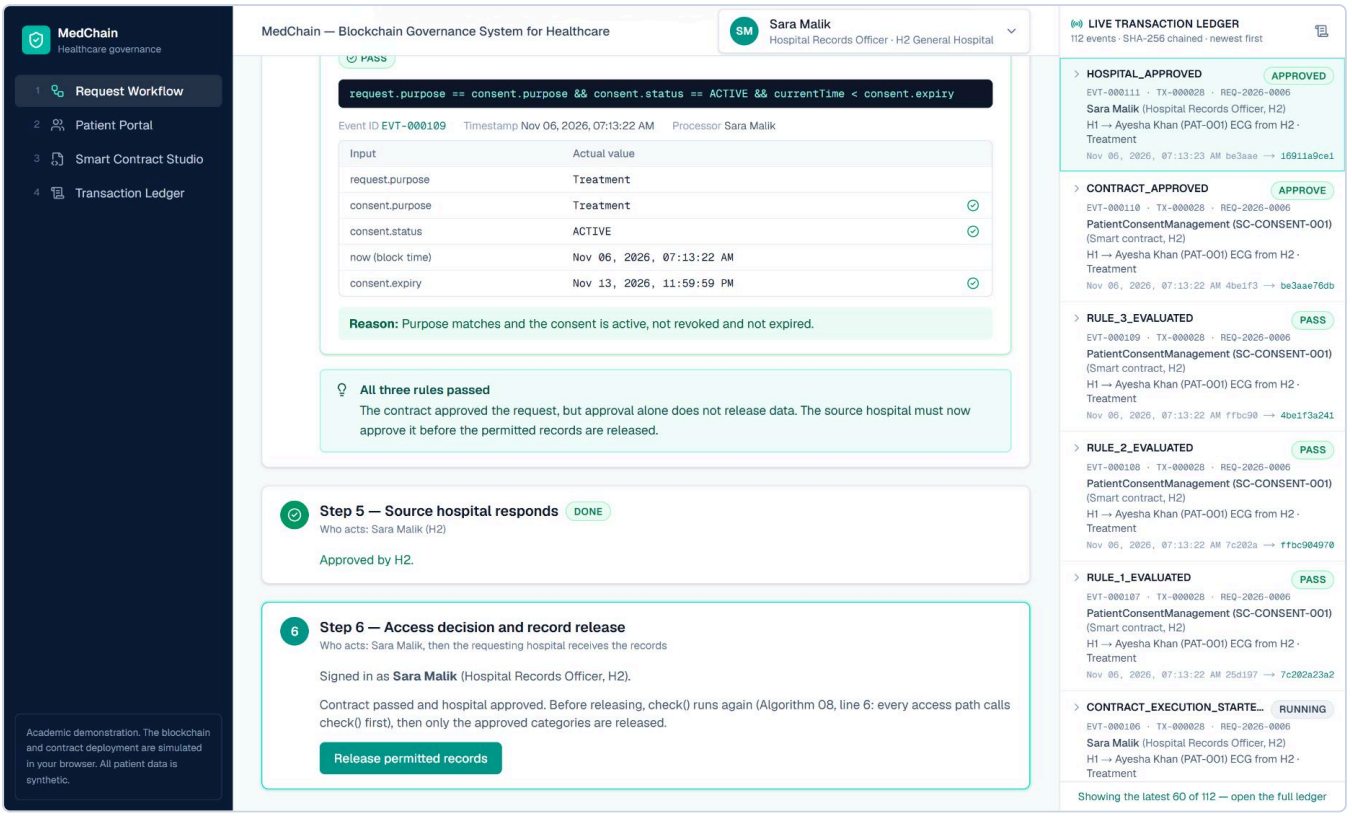


Figure 20. Contract approved and hospital approved. No record has been released yet.

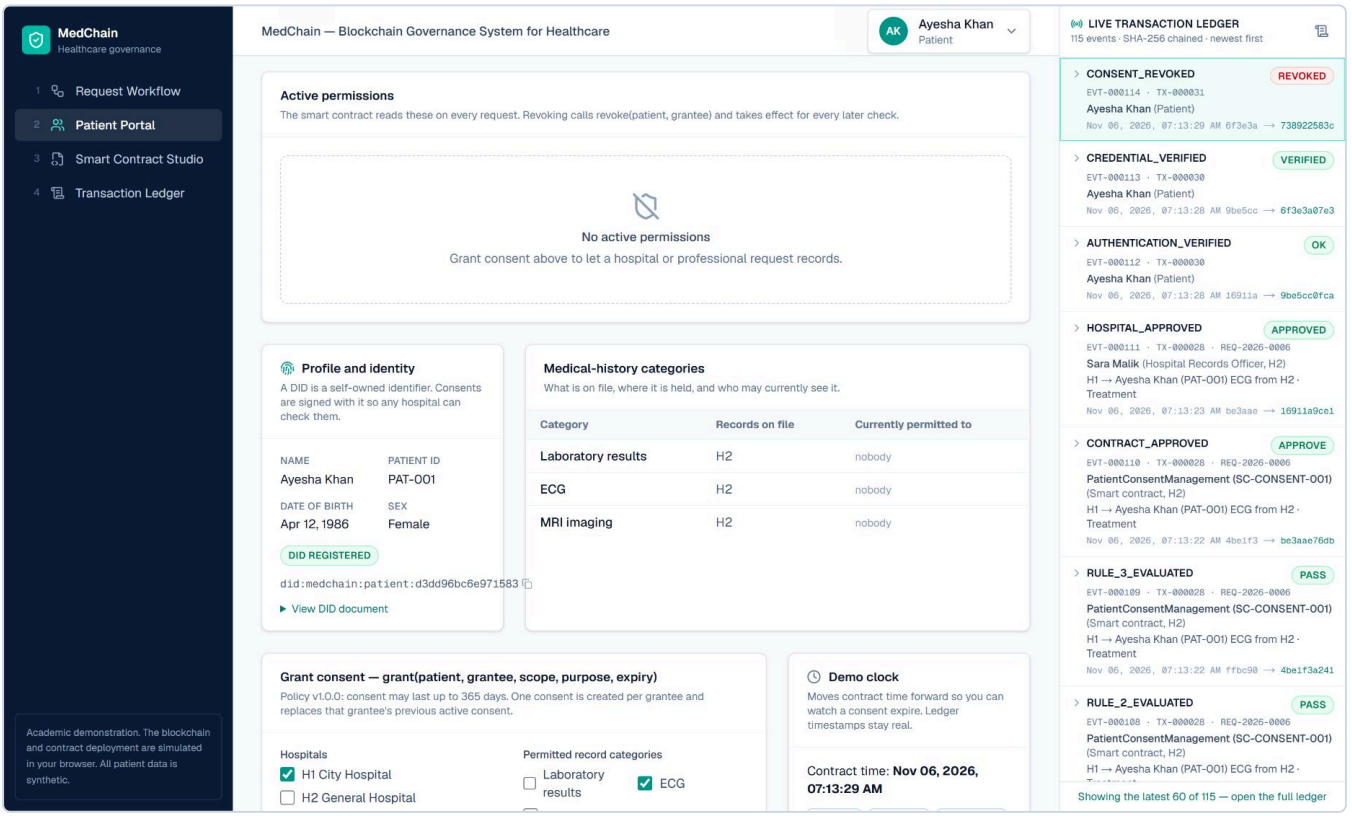


Figure 21. Ayesha revokes City Hospital: revoke(p, g). CONSENT_REVOKED appears in the live ledger.

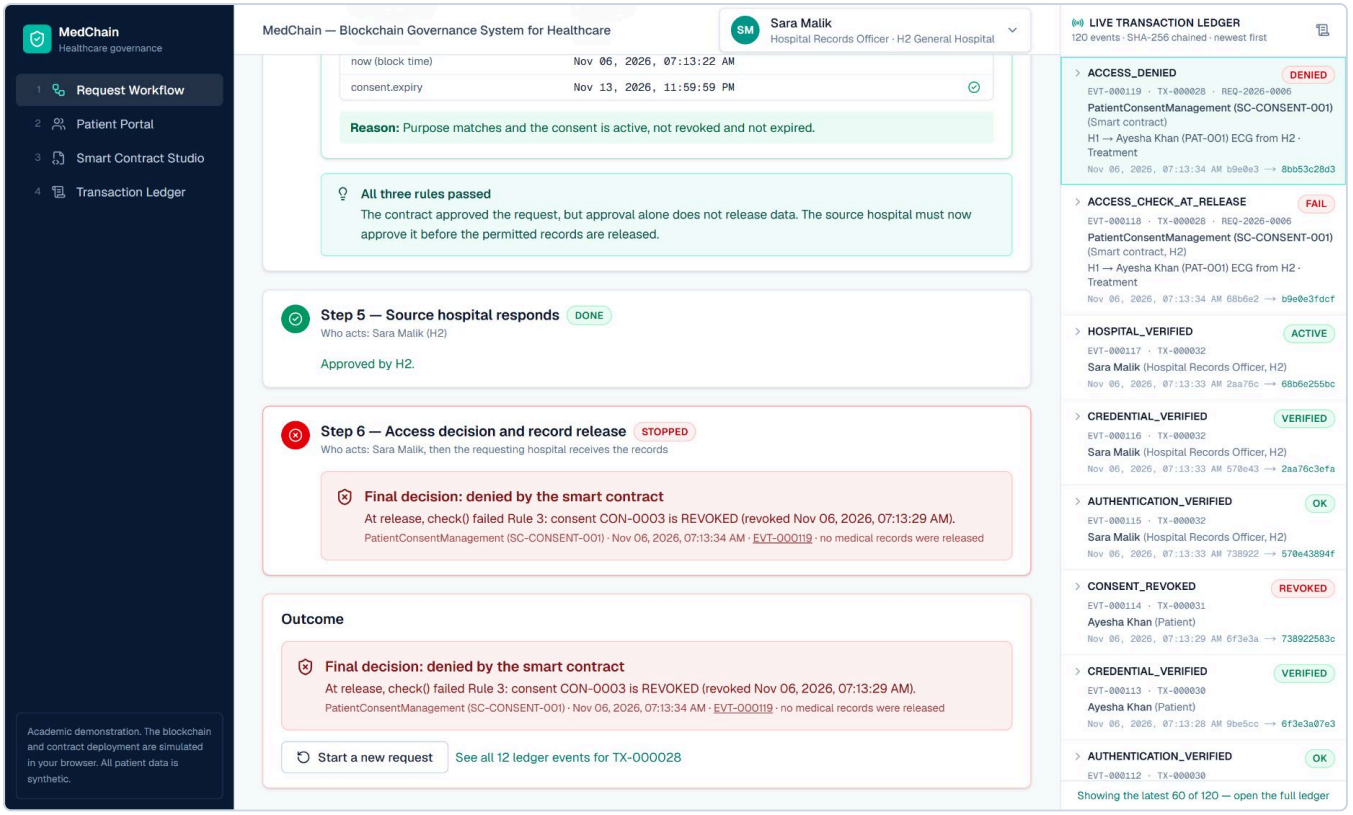


Figure 22. Release re-runs check(): Rule 3 fails (consent REVOKED). ACCESS_CHECK_AT_RELEASE FAIL and ACCESS_DENIED are recorded, and nothing is released.

Events written by the release attempt

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000118 TX-000028 REQ-2026-0006	**ACCESS_CHECK_AT_RELEASE** check() re-run at the moment of access failed Rule 3: consent CON-0003 is REVOKED (revoked Nov 06, 2026, 07:13:29 AM).	PatientConsentManagement (SC-CONSENT-001) Smart contract, H2	FAIL	68b6e255bc51… ↓ **b9e0e3fdcf28…**
EVT-000119 TX-000028 REQ-2026-0006	**ACCESS_DENIED** Access denied. At release, check() failed Rule 3: consent CON-0003 is REVOKED (revoked Nov 06, 2026, 07:13:29 AM). No medical records were released.	PatientConsentManagement (SC-CONSENT-001) Smart contract	DENIED	b9e0e3fdcf28… ↓ **8bb53c28d3d3…**

9. Patient Portal — consent and access history

The patient can see their profile, record categories and who may currently see each one; grant consent (to hospitals or designated professionals; purpose Treatment or Research; expiry); see active permissions and revoke them; and see every request for their records with the requesting hospital, the decision and the reason.

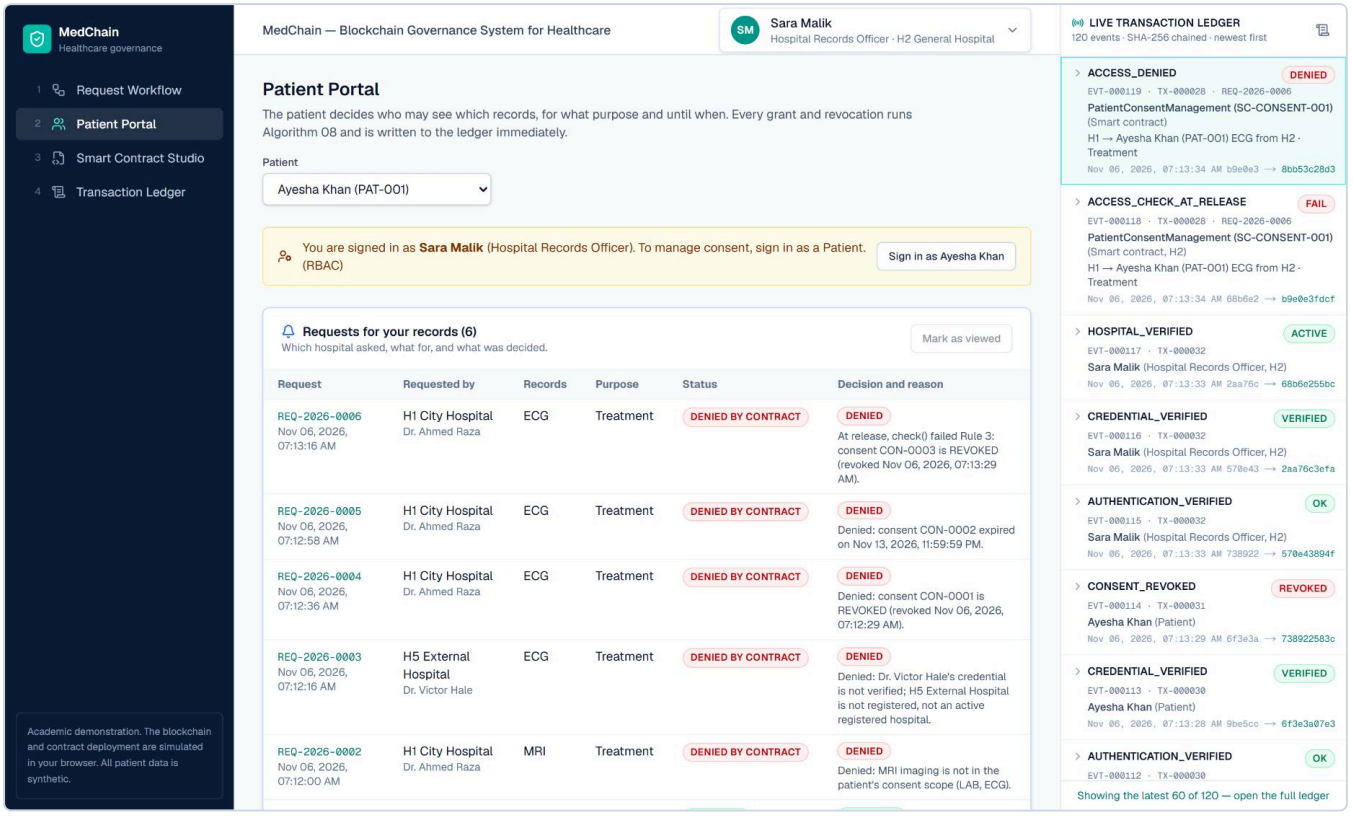


Figure 23. Patient Portal: all requests for Ayesha's records with status, decision and reason.

10. Ledger integrity — verification, tampering and hash inspection

Verification logic: for every event in order, check that its previousHash equals the stored hash of the event before it, then recompute SHA256(previousHash + canonicalEventPayload) and compare it with the stored currentHash. The first mismatch is reported with its position.

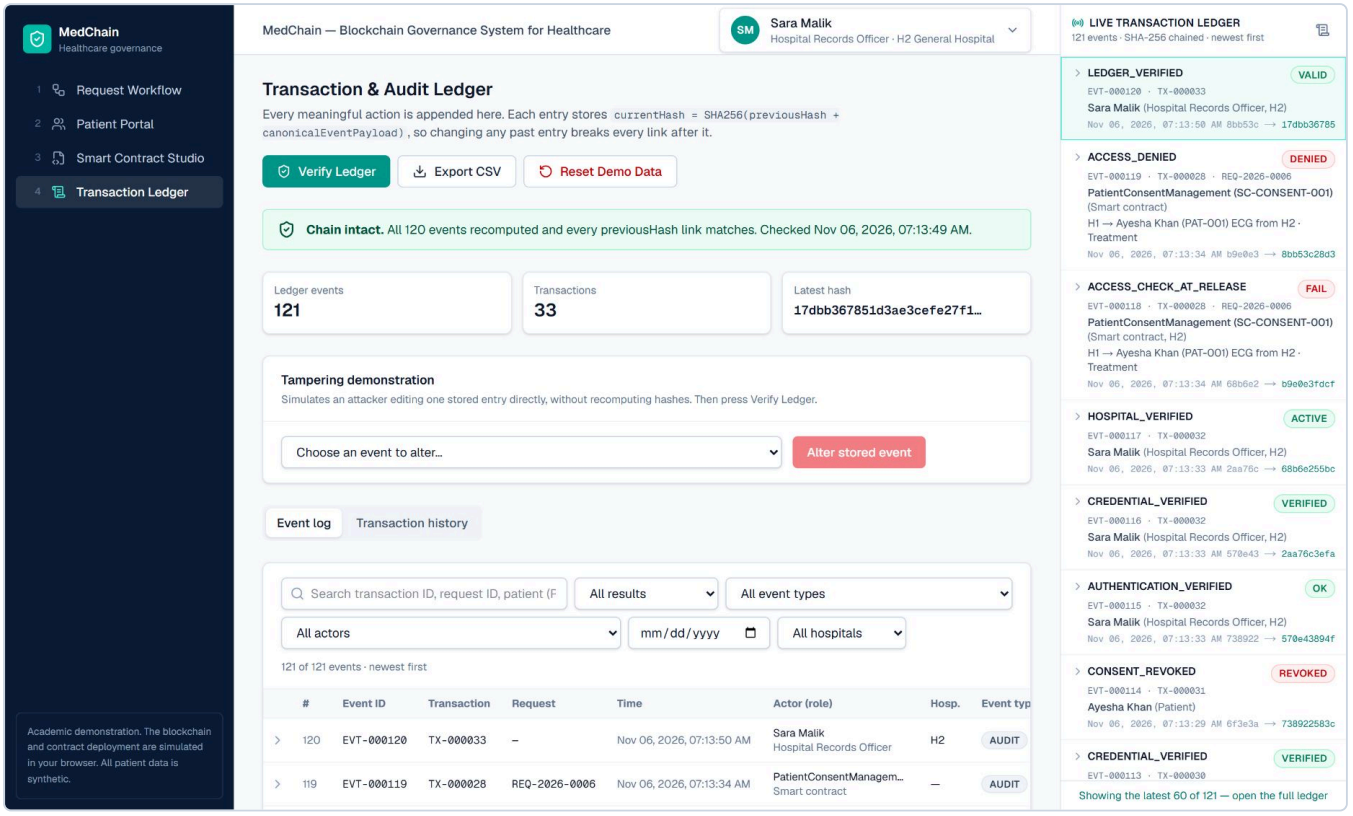


Figure 24. Verify Ledger: chain intact; every event recomputed.

Tampering demonstration: the tool edits one stored event directly, the way an attacker editing storage would, without recomputing any hash and without writing a ledger event.

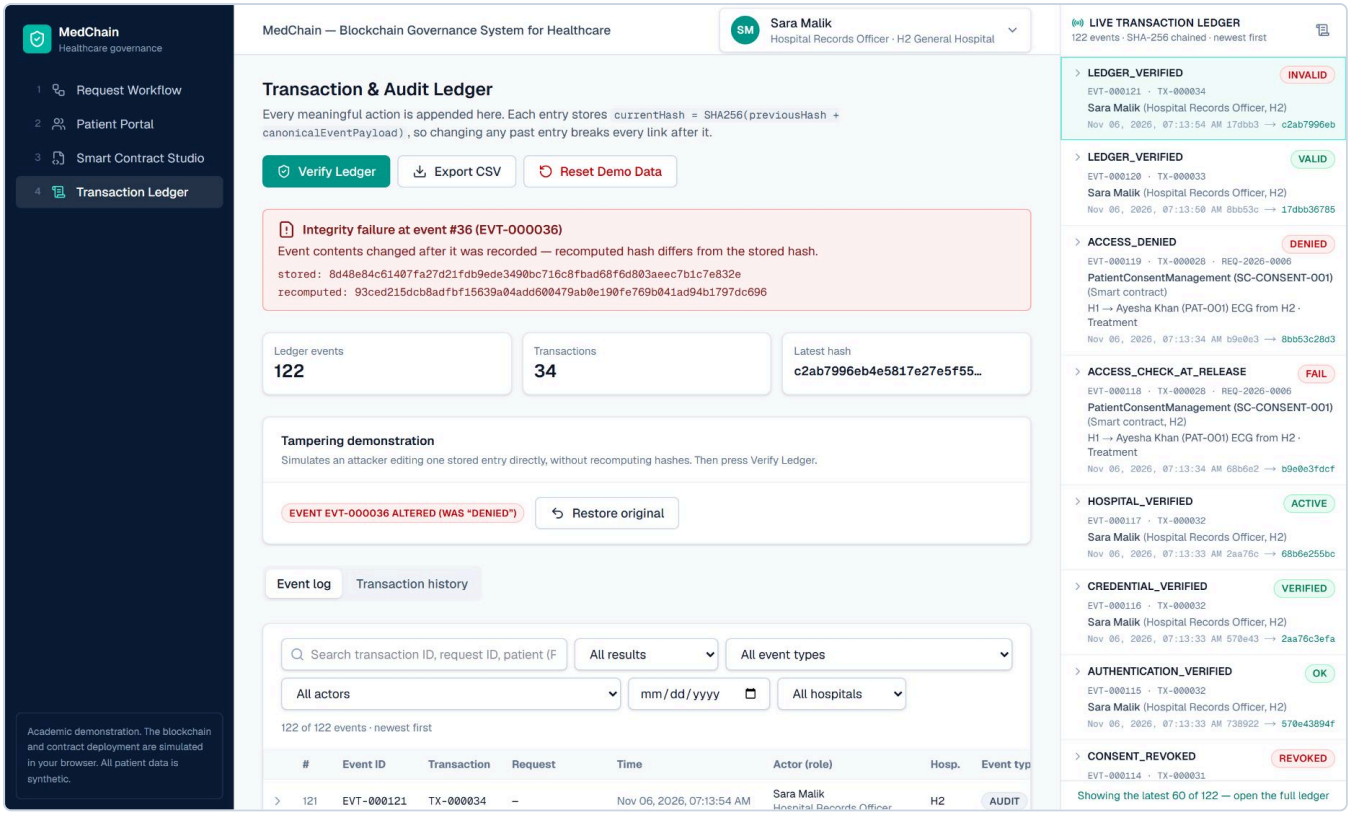


Figure 25. An ACCESS_DENIED event was changed to GRANTED in storage. Verification fails at exactly that event and shows the stored hash next to the recomputed one.

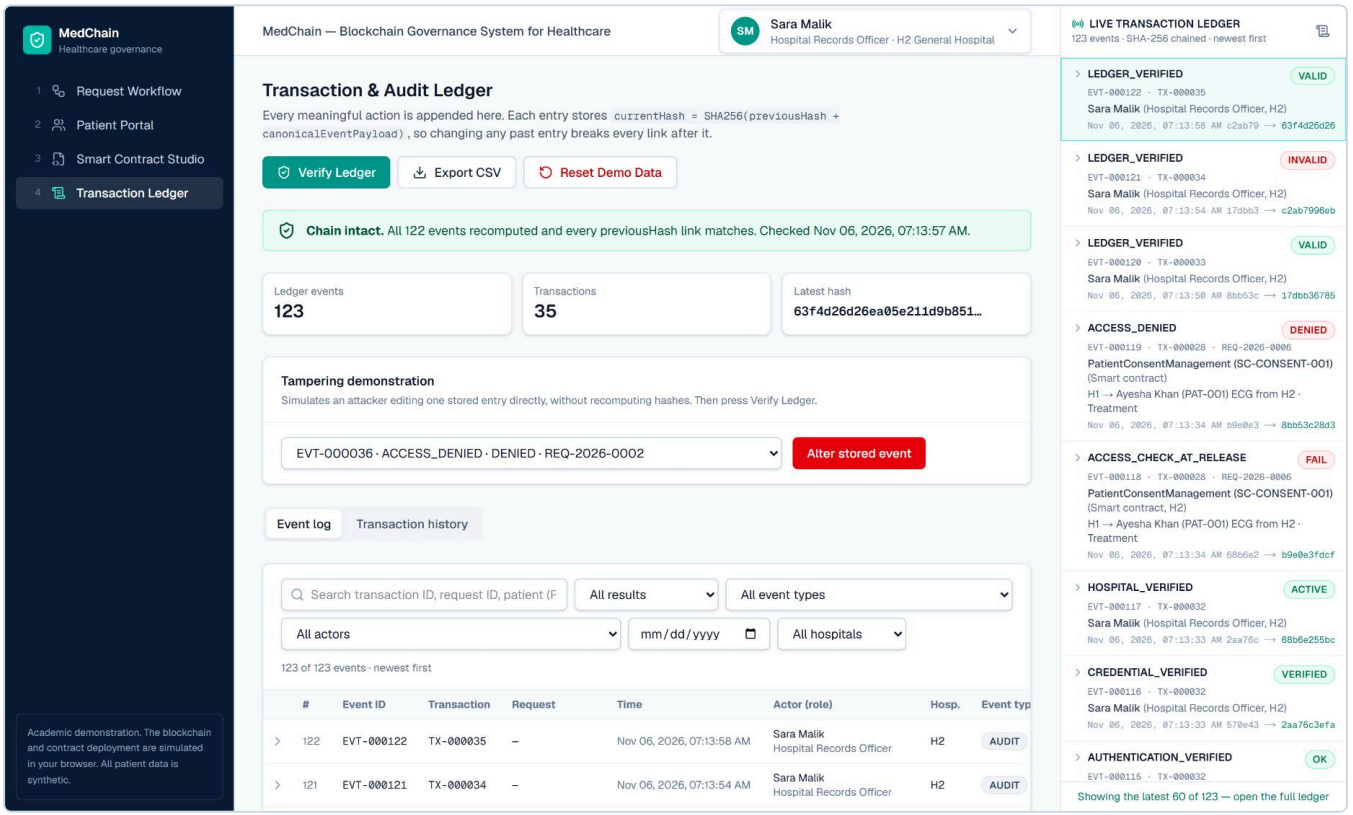


Figure 26. After 'Restore original', verification passes again.

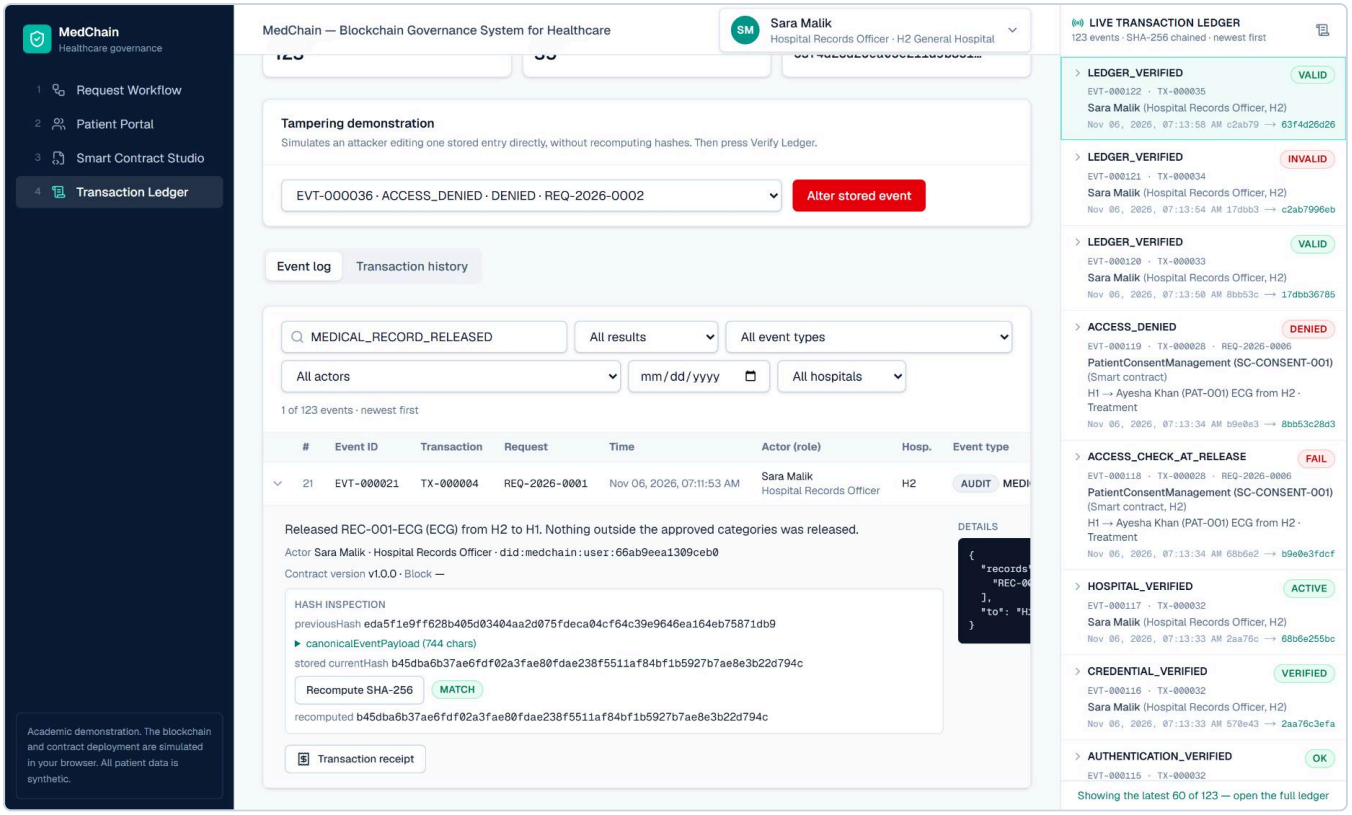


Figure 27. Hash inspection for one event: previousHash, the canonical payload, the stored hash and the hash recomputed live with SHA-256 — they match.

The ledger screen also provides search by transaction ID, request ID or patient; filters by event type, actor, hospital, date and outcome; a transaction-history view grouping events by transaction ID; a transaction receipt per event; and CSV export of all events with every field.

11. Smart Contract Studio

Contract: PatientConsentManagement — ID **SC-CONSENT-001**, version **1.0.0**, SHA-256 code hash `136ca39a89d15f4304153eb7e30a0184d1c019fce3f6f2fc43737fa7e1c68b5e`. Deployment is simulated and produces a receipt (transaction, block, deployer, simulated address).

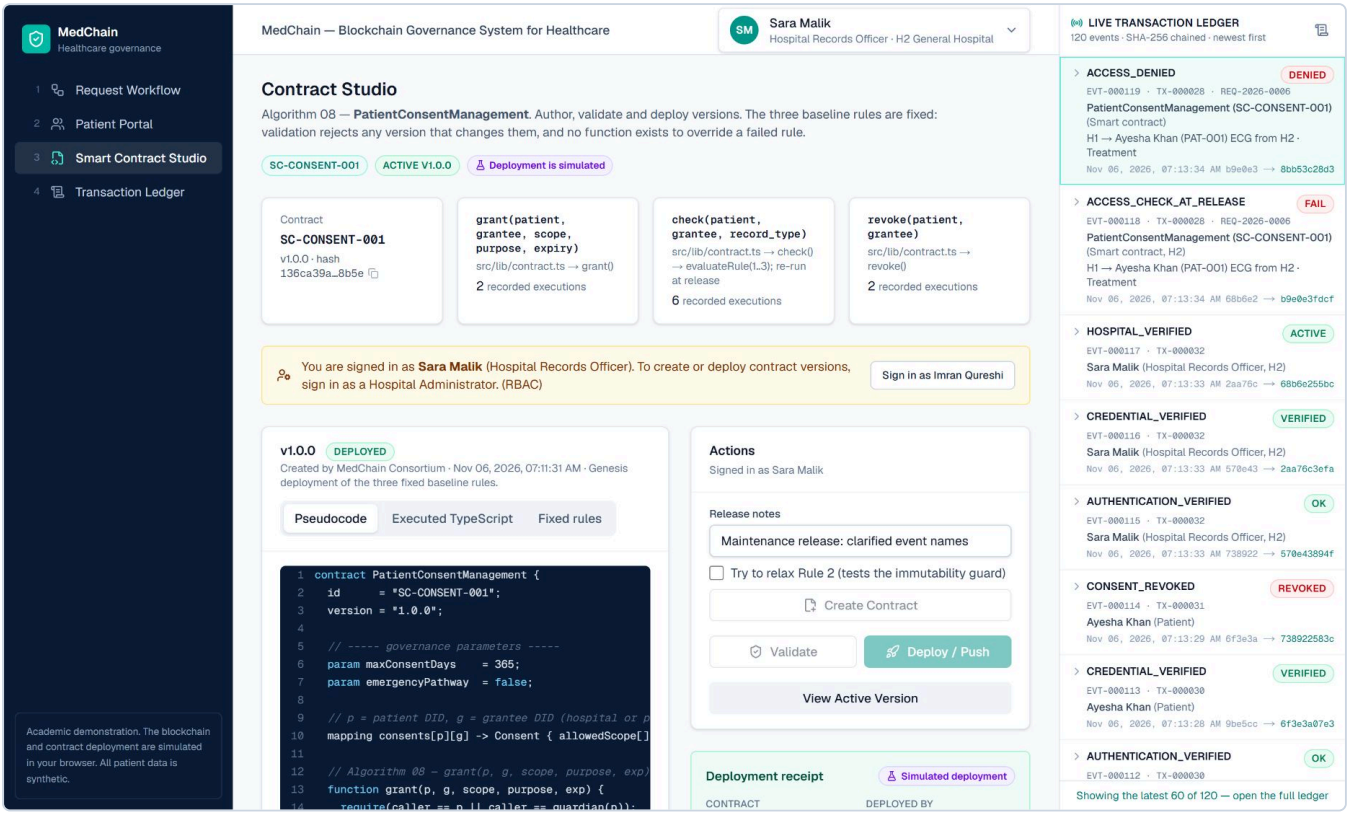


Figure 28. Contract ID, active version and hash; pseudocode with grant(), check(), revoke(); the shaded lines are the fixed rules.

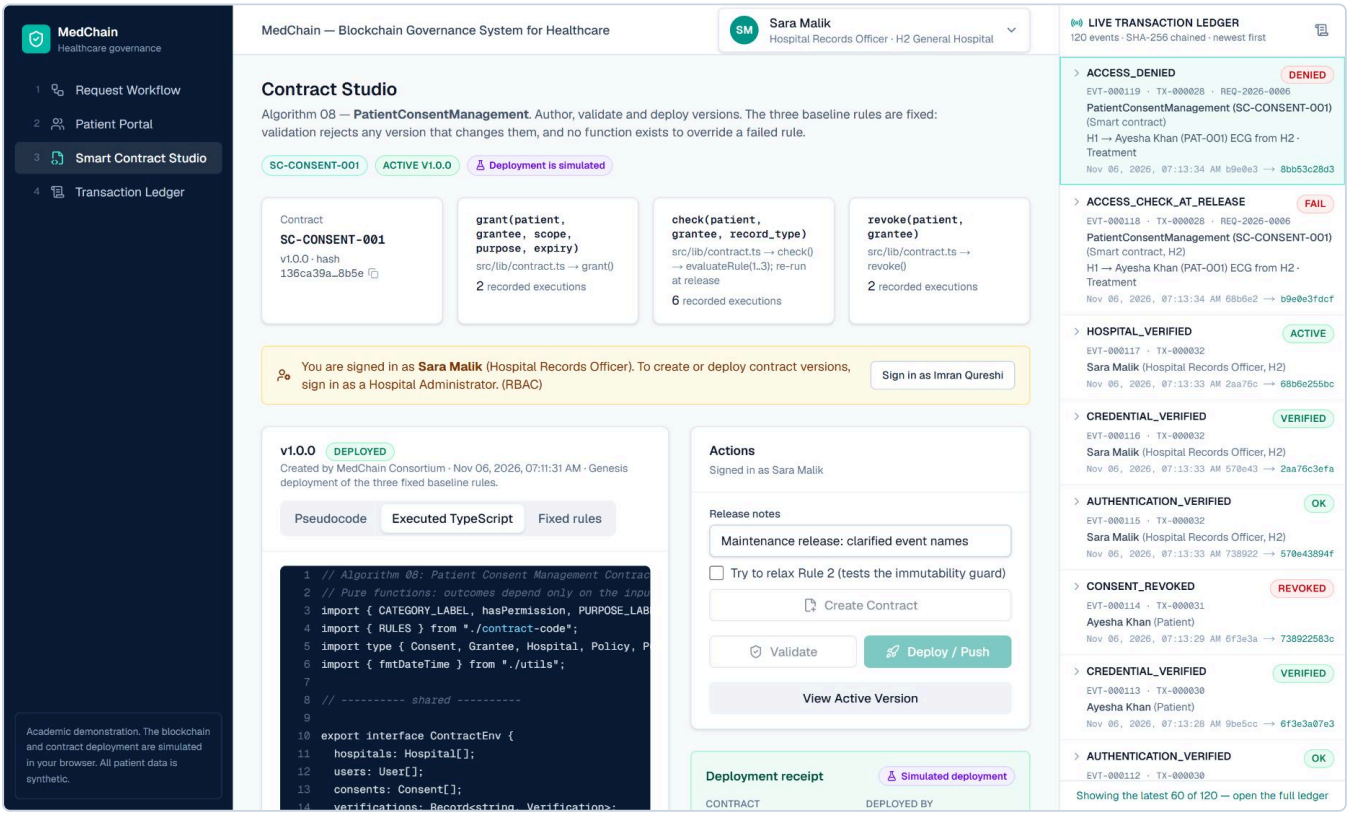


Figure 29. The TypeScript the application actually executes (src/lib/contract.ts), served from the source file, with its SHA-256 hash, so the displayed algorithm matches the executed code.

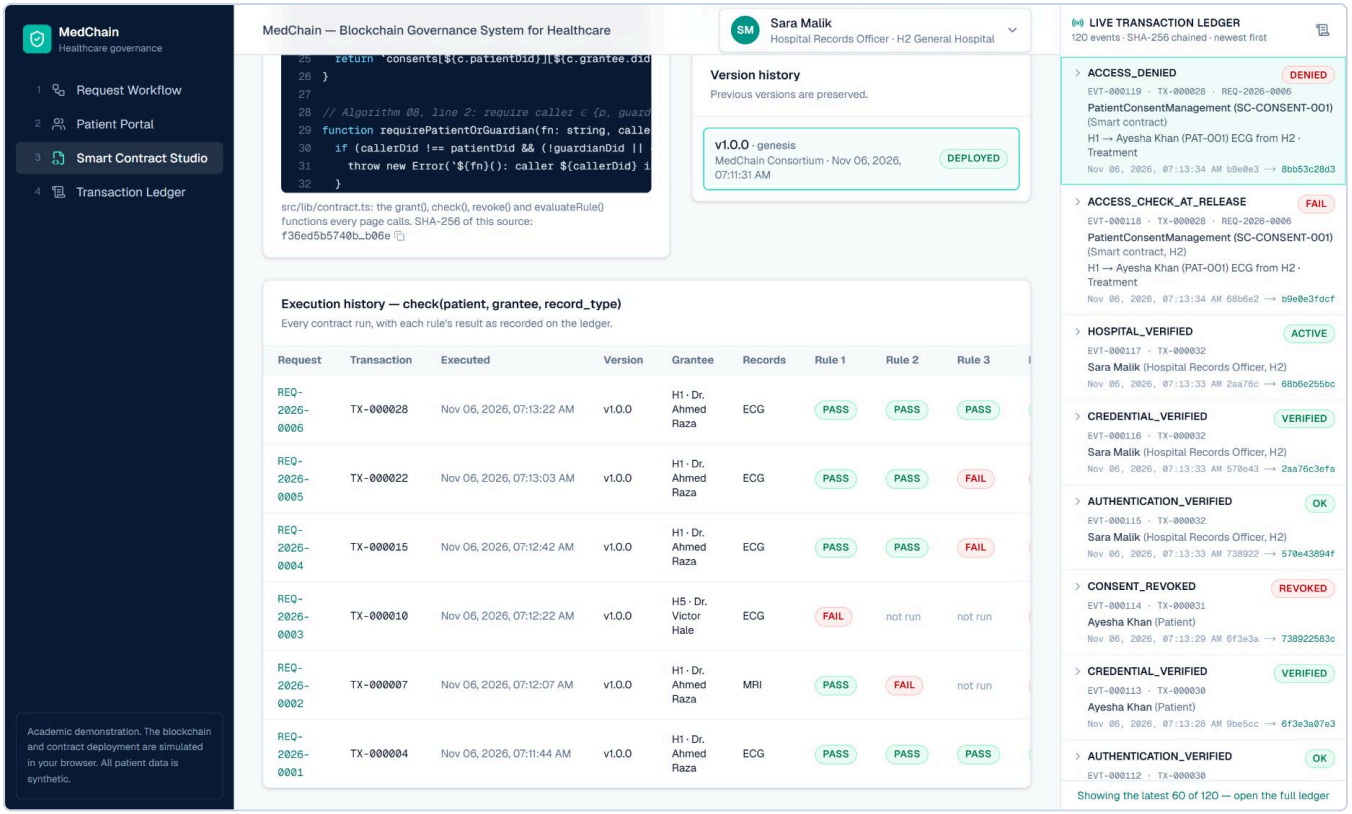


Figure 30. Execution history of check() for every request: result of each rule and the decision.

12. Requirement compliance matrix

Requirement	How it is met	Evidence
Algorithm 08 implemented as a reusable TypeScript module with grant(), check(), revoke()	src/lib/contract.ts; used by every page through the central store	§4, Fig. (Contract Studio)
Three fixed rules R1–R3 evaluated individually with inputs, PASS/FAIL and explanation	evaluateRule(1..3); each result stored and shown with its event ID	§6 Step 4
Any failed rule denies access; no override	check() stops at the first failure; the UI offers no approve/release action after a denial; the store rejects approval unless status is CONTRACT_APPROVED	§7, §8
Step 1 — authentication, credential, hospital, role	Sign-in records AUTHENTICATION_VERIFIED, CREDENTIAL_VERIFIED, HOSPITAL_VERIFIED	§6 Step 1
Step 2 — patient history request with request ID and transaction reference	REQUEST_CREATED; e.g. REQ-2026-0001 / TX-000004	§6 Step 2
Step 3 — source hospital receives and reviews	REQUEST_RECEIVED, REQUEST_REVIEW_STARTED	§6 Step 3
Step 4 — hospital verification and response (approve or reject with reason)	HOSPITAL_VERIFICATION_COMPLETED; HOSPITAL_APPROVED or HOSPITAL_REJECTED	§6 Steps 3 and 5
Step 5 — smart-contract execution	CONTRACT_EXECUTION_STARTED, RULE_1/2/3_EVALUATED, CONTRACT_APPROVED / CONTRACT_DENIED	§6 Step 4, §7
Step 6 — final decision; records released only if rules pass and hospital approves	ACCESS_GRANTED + MEDICAL_RECORD_RELEASED, or ACCESS_DENIED; only requested and consented categories released	§6 Step 6
Step 7 — patient notification and access history	Patient Portal lists every request, hospital, decision and reason	§9
Patient Portal: profile, categories, grant, purpose (Treatment/Research), expiry, active permissions, revoke, history	Patient Portal screen	§8, §9
Revocation immediately affects later checks	Revoked consent → REQ-2026-0004 denied by Rule 3	§8.2
Lecture line 2: require caller ∈ {p, guardian(p)}	requirePatientOrGuardian() in grant() and revoke(); caller DID compared with the patient and guardian DIDs	§4.1
Lecture line 3: consents[p][g] keyed by DIDs	Consent stores patientDid and grantee DID; key shown on screen and in ledger events	§4.1
Lecture line 6: every access path calls check() first	check() re-run at release; REQ-2026-0006 blocked after a late revocation	§4.4, §8.4
Every meaningful activity recorded; Event ID, Transaction ID, Request ID, timestamp, actor and role, hospital, event type, result, explanation, previous and current hash	recordLedgerEvent() in src/lib/store.tsx	§5, all event tables
currentHash = SHA256(previousHash + payload), chained to the previous event	Web Crypto API; canonical JSON payload; genesis previousHash = 64 zeros	§5, Appendix A
Live ledger updates, transaction history, event details, hash inspection, verification, tampering demo, CSV export	Right-side live ledger; Transaction Ledger screen	§10
Tampering detected	Altered event detected at its exact position	§10
Smart Contract Studio: algorithm, rules, functions, ID, version, hash, execution history, simulated deployment	SC-CONSENT-001 v1.0.0, hash 136ca39a89d15f43…	§11

Requirement	How it is met	Evidence
Roles: Patient, Physician, Records Officer, Hospital Administrator, Auditor	One demo user per role (table in §3)	§3
Sample data: Ayesha Khan (PAT-001); H1–H5; consent H1, LAB+ECG, not MRI, Treatment, 31 Dec 2026, ACTIVE	Seed data, shown in the Patient Portal and in SEED_CONSENT_LOADED	§3
Scenarios 1–6 executed by the real algorithm	Approved ECG; MRI (R2); H5 (R1); revoked (R3); expired (R3); tampering	§6–§10
Next.js, TypeScript, Tailwind, central state, localStorage, Web Crypto SHA-256	Next.js 16 App Router; React Context store; persisted after each committed action	§2

13. Test results summary

Scenario	Request	Rules	Outcome	Records released
1. Successful ECG request	`REQ-2026-0001` `TX-000004`	R1 PASS, R2 PASS, R3 PASS	RELEASED All three contract rules passed, the source hospital approved, and check() passed again at release.	REC-001-ECG
2. MRI outside consent	`REQ-2026-0002` `TX-000007`	R1 PASS, R2 FAIL	DENIED Denied: MRI imaging is not in the patient's consent scope (LAB, ECG).	none
3. Unauthorized hospital H5	`REQ-2026-0003` `TX-000010`	R1 FAIL	DENIED Denied: Dr. Victor Hale's credential is not verified; H5 External Hospital is not registered, not an active registered hospital.	none
4. Revoked consent	`REQ-2026-0004` `TX-000015`	R1 PASS, R2 PASS, R3 FAIL	DENIED Denied: consent CON-0001 is REVOKED (revoked Nov 06, 2026, 07:12:29 AM).	none
5. Expired consent	`REQ-2026-0005` `TX-000022`	R1 PASS, R2 PASS, R3 FAIL	DENIED Denied: consent CON-0002 expired on Nov 13, 2026, 11:59:59 PM.	none
5b. Consent revoked after approval	`REQ-2026-0006` `TX-000028`	R1 PASS, R2 PASS, R3 PASS; at release R3 FAIL	DENIED At release, check() failed Rule 3: consent CON-0003 is REVOKED (revoked Nov 06, 2026, 07:13:29 AM).	none
6. Ledger tampering	—	—	Verification failed at the altered event; passed again after restore	—

All outcomes above come from the same algorithm, run on the data at the time of each request. No outcome is hard-coded.

Limitations (simulated parts)

- The ledger runs in one browser and is persisted in localStorage; there is no distributed network.
- Credential "signatures" are SHA-256 proofs over the credential content, standing in for real digital signatures.
- Contract deployment is simulated and produces a receipt but no real on-chain transaction.

Appendix A — Full hash chain of the approved transaction (TX-000004)

Each event's previousHash is the currentHash of the event immediately before it in the whole ledger, so the chain links across transactions as well.

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000006 TX-000004 REQ-2026-0001	**REQUEST_CREATED** Dr. Ahmed Raza (H1) requested Ayesha Khan's ECG from H2 for Treatment. Reason: Follow-up of recent chest pain; need prior ECG for comparison.	Dr. Ahmed Raza Physician, H1	SUBMITTED	ccfd6acefa54fae30cc9b878d130c4949216a9aed04465778d323eecf11f8b6f ↓ **19f36408ffbb41450e3cf1ea3102c9a257cb082d54db75f925dfb673f8d52eae**
EVT-000010 TX-000004 REQ-2026-0001	**REQUEST_RECEIVED** H2 received REQ-2026-0001 in its inbox; Sara Malik acknowledged it.	Sara Malik Hospital Records Officer, H2	RECEIVED	c3dc20100a79dba587e519a54d0b14318bfe493863e0eee7ecfbed3e3b8e344e ↓ **cae8af7a82caa6ac3ad010c423d6d69cd74898ecee94933fd2c82e3295cfa248**
EVT-000011 TX-000004 REQ-2026-0001	**REQUEST_REVIEW_STARTED** Sara Malik opened REQ-2026-0001 and started reviewing it.	Sara Malik Hospital Records Officer, H2	UNDER_REVIEW	cae8af7a82caa6ac3ad010c423d6d69cd74898ecee94933fd2c82e3295cfa248 ↓ **5e1b4a633ecc044318ae6f6392adb3a178da63fd7fab958e4486f1d77e4b296e**
EVT-000012 TX-000004 REQ-2026-0001	**HOSPITAL_VERIFICATION_COMPLETED** Sara Malik verified REQ-2026-0001: patient reference OK, requesting hospital OK, requesting professional OK, requested data held here OK, patient consent on file OK. This is the hospital's check; Algorithm 08 still decides independently.	Sara Malik Hospital Records Officer, H2	VERIFIED	5e1b4a633ecc044318ae6f6392adb3a178da63fd7fab958e4486f1d77e4b296e ↓ **14fbf2aee34eb9810fcb229c7a59366c406c3d978e26d4f653d3989702e04570**
EVT-000013 TX-000004 REQ-2026-0001	**CONTRACT_EXECUTION_STARTED** Sara Malik started PatientConsentManagement v1.0.0: check(PAT-001, u-phy-h1@H1, [ECG]).	Sara Malik Hospital Records Officer, H2	RUNNING	14fbf2aee34eb9810fcb229c7a59366c406c3d978e26d4f653d3989702e04570 ↓ **913afb9e8a76f2d4512aedb2a69b5545f8dbe0a82c33927fdee844427105cd19**
EVT-000014 TX-000004 REQ-2026-0001	**RULE_1_EVALUATED** Rule 1 (Identity and hospital authorization): Both professionals are signed in with verified credentials, work at active registered hospitals, and hold the required role permissions.	PatientConsent Management (SC-CONSENT-001) Smart contract, H2	PASS	913afb9e8a76f2d4512aedb2a69b5545f8dbe0a82c33927fdee844427105cd19 ↓ **9fe10d6b1e7f2448315ff1b47709fb8f20f8339fc8ea19d41ebf6eb12bdaea8c**
EVT-000015 TX-000004 REQ-2026-0001	**RULE_2_EVALUATED** Rule 2 (Patient consent and record scope): Consent CON-0001 authorizes H1 City Hospital and covers all requested types (ECG).	PatientConsent Management (SC-CONSENT-001) Smart contract, H2	PASS	9fe10d6b1e7f2448315ff1b47709fb8f20f8339fc8ea19d41ebf6eb12bdaea8c ↓ **83ee1fbed31bbe7675d0aaf3a56dea5c871f2c858b94e5f94b6b28f1392fd549**
EVT-000016 TX-000004 REQ-2026-0001	**RULE_3_EVALUATED** Rule 3 (Purpose, expiry and revocation): Purpose matches and the consent is active, not revoked and not expired.	PatientConsent Management (SC-CONSENT-001) Smart contract, H2	PASS	83ee1fbed31bbe7675d0aaf3a56dea5c871f2c858b94e5f94b6b28f1392fd549 ↓ **27170f4014e6e72699402e355a14643797d7c7e43e1689d5d7e412d8b653e5d8**
EVT-000017 TX-000004 REQ-2026-0001	**CONTRACT_APPROVED** check() returned APPROVE: all three fixed rules passed. The source hospital must now approve or reject.	PatientConsent Management (SC-CONSENT-001) Smart contract, H2	APPROVE	27170f4014e6e72699402e355a14643797d7c7e43e1689d5d7e412d8b653e5d8 ↓ **cf65f5ccbebd439c8761ddbd64a3f8cf17c2c931bfbaa9ffbb41969e4abbf912**

Event / Tx / Request	Event type and explanation (as recorded)	Actor (role)	Result	previousHash → currentHash
EVT-000018 TX-000004 REQ-2026-0001	**HOSPITAL_APPROVED** Sara Malik (H2) approved REQ-2026-0001. Contract passed and hospital approved: the permitted records may now be released.	Sara Malik Hospital Records Officer, H2	APPROVED	cf65f5ccbebd439c8761ddbd64a3f8cf17c2c931bfbaa9ffbb41969e4abbf912 ↓ **f3eea2de2de50c3465a2da7b451458d27baf10b88becb684daefd2522aa99976**
EVT-000019 TX-000004 REQ-2026-0001	**ACCESS_CHECK_AT_RELEASE** check() re-run at the moment of access: Rules 1–3 still pass, so the release may go ahead.	PatientConsent Management (SC-CONSENT-001) Smart contract, H2	PASS	f3eea2de2de50c3465a2da7b451458d27baf10b88becb684daefd2522aa99976 ↓ **65a5cac53ca1bf20b91dd8d267a58ae37264fb2fcbae752a4b374a5697b441ed**
EVT-000020 TX-000004 REQ-2026-0001	**ACCESS_GRANTED** Access granted: all three contract rules passed and the source hospital approved.	Sara Malik Hospital Records Officer, H2	GRANTED	65a5cac53ca1bf20b91dd8d267a58ae37264fb2fcbae752a4b374a5697b441ed ↓ **eda5f1e9ff628b405d03404aa2d075fdeca04cf64c39e9646ea164eb75871db9**
EVT-000021 TX-000004 REQ-2026-0001	**MEDICAL_RECORD_RELEASED** Released REC-001-ECG (ECG) from H2 to H1. Nothing outside the approved categories was released.	Sara Malik Hospital Records Officer, H2	RELEASED	eda5f1e9ff628b405d03404aa2d075fdeca04cf64c39e9646ea164eb75871db9 ↓ **b45dba6b37ae6fdf02a3fae80fdae238f5511af84bf1b5927b7ae8e3b22d794c**

Appendix B — Contract code as deployed (SC-CONSENT-001 v1.0.0)

```
contract PatientConsentManagement {
  id      = "SC-CONSENT-001";
  version = "1.0.0";

  // ----- governance parameters -----
  param maxConsentDays    = 365;
  param emergencyPathway  = false;

  // p = patient DID, g = grantee DID (hospital or professional), now = block time
  mapping consents[p][g] -> Consent { allowedScope[]; purpose; expiry; status; }

  // Algorithm 08 — grant(p, g, scope, purpose, exp)
  function grant(p, g, scope, purpose, exp) {
    require(caller == p || caller == guardian(p));                    // line 2
    require(did.isRegistered(p) && scope.length > 0);
    require(now < exp && exp <= now + days(maxConsentDays));
    consents[p][g] = Consent(scope, purpose, exp, ACTIVE);            // line 3 (replaces the previous entry)
    emit ConsentGranted(p, g, scope, purpose, exp);
  }

  // Algorithm 08 — check(p, g, type): type in scope && now < exp     // line 4
  // g = the requesting professional and their hospital; the request supplies purpose and processor.
  function check(p, g, type) returns (Decision) {
    consent = consents[p][g];                                         // newest ACTIVE entry, else the latest one
    if (!R1_IdentityAndHospital(g, processor))                 return deny(R1);
    if (!R2_ConsentAndScope(consent, g, type))                 return deny(R2);   // type in scope
    if (!R3_PurposeExpiryRevocation(consent, request.purpose, now)) return deny(R3); // now < exp, not revoked
    emit ContractDecision(request.id, APPROVE);
    return APPROVE;
  }

  // Algorithm 08 — revoke(p, g)                                      // line 5
  function revoke(p, g) {
    require(caller == p || caller == guardian(p));
    consents[p][g].status = REVOKED;   // equivalent to delete for access; kept so Rule 3 can say "revoked" and the
audit trail stays complete
    emit ConsentRevoked(p, g, now);
  }

  // Algorithm 08 — line 6: every access path calls check() first
  function releaseRecords(request) {
    require(request.contractDecision == APPROVE && request.hospitalApproved);
    require(check(request.patient, request.grantee, request.recordTypes) == APPROVE);   // re-checked at the moment
of access
    release(records where type in request.recordTypes);
  }

  // A failed rule always denies. There is no override function.

  // ===== FIXED BASELINE RULES — immutable, no user or administrator can edit or override them =====
  rule R1_IdentityAndHospital(grantee, processor) returns (bool) {
    return grantee.authenticated && credentials[grantee].verified
        && hospitals[grantee.hospital].status == ACTIVE
        && RBAC.has(grantee.role, "request:create")
        && processor.authenticated && credentials[processor].verified
        && RBAC.has(processor.role, "request:process")
        && hospitals[request.sourceHospital].status == ACTIVE
        && processor.hospital == request.sourceHospital;
  }

  rule R2_ConsentAndScope(consent, grantee, record_type) returns (bool) {
    return consent != null
        && (consent.grantee == grantee.hospital || consent.grantee == grantee)
        && all(t IN consent.allowedScope for t in record_type);
  }

  rule R3_PurposeExpiryRevocation(consent, purpose, now) returns (bool) {
    return purpose == consent.purpose
```

```
        && consent.status == ACTIVE        // not revoked or superseded
        && now < consent.expiry;           // not expired
  }
}
```